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Introduction to Dark Matter and Cosmology

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ABSTRACT: Lecture notes for the 2026 International Summer School for Top Physics Students, Beihang University, China.

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1 Introduction

Two statements summarize our understanding of dark matter today: (1) dark matter exists, and (2) its nature remains elusive. There is a wealth of textbooks, reviews, community reports, lecture notes, and video recordings on the subject of dark matter. Notable recent contributions include the review by [1], the textbooks by [2, 3], the TASI lectures by [4–6], on which I heavily relied in preparing these lecture notes. Given the vast scope of dark matter modeling and detection, it is impossible to cover all aspects comprehensively in these lectures. Therefore, we will focus on an emerging paradigm I have been working on, where dark matter is considered part of a dark sector. In this framework, additional particles in the dark sector serve as mediators for interactions between dark matter and ordinary matter, and for interactions among dark matter particles themselves.

The notes are organized as follows. The Pre-lecture (Sec. 2) collects the background that the lectures rely on: natural units, the art of estimation, and a minimal toolkit of FRW cosmology. Lecture I (Sec. 3) reviews the evidence for dark matter and its established properties, introduces thermal freeze-out and the WIMP paradigm, and then turns to dark sectors: portal interactions, the complementarity of intensity-frontier, astrophysical, and cosmological probes of a muon-philic dark scalar, and the N_{eff} constraints from BBN and the CMB. Lecture II (Sec. 4) begins in the early universe, deriving the scale-invariant primordial power spectrum from inflation, and follows it to late times: the linear matter power spectrum, its small-scale suppression in beyond- Λ CDM models, and the observations that probe it. Lecture III (Sec. 5) zooms into individual halos: spherical collapse and the splashback radius, the Press-Schechter halo mass function, and the gravothermal evolution of self-interacting dark matter halos. Topics that we do not cover include the experimental techniques of direct detection [4], axion and ultralight dark matter phenomenology [6, 7], and indirect and collider searches [1]. Throughout the notes, boxed *Questions* come with worked answers, while boxed *Exercises* are left for the reader. Subsections marked with a star (★) contain more advanced material and can be skipped on a first reading.

Before concluding the introduction, I would like to share my favorite quote:

“It is difficult but interesting to master ten percent of any field. The path from ten to ninety percent is pure pleasure and genuine creativity. To go through the next nine percent is infinitely difficult, and far from everyone’s ability. The last percent is hopeless. It is more reasonable to switch to a new problem before it is too late.”

— Yakov Zel’dovich

2 Pre-lecture

2.1 Natural units

Throughout these notes, we work in natural units, assuming

$$c = \hbar = k_B = \epsilon_0 = \mu_0 = 1. \quad (2.1)$$

Setting $c = 1$ ties lengths to times, $\hbar = 1$ ties energies to inverse times, and $k_B = 1$ ties temperatures to energies. As a result, every dimensionful quantity can be expressed in powers of a single unit, conventionally an energy (eV or GeV):

$$[\text{mass}] = [\text{temperature}] = [\text{energy}] = [\text{length}]^{-1} = [\text{time}]^{-1}. \quad (2.2)$$

Two conversion factors, essentially $\hbar c$ and $\hbar$, are enough to translate lengths and times back to CGS/SI units,

$$1 \text{ GeV}^{-1} = 1.97 \times 10^{-14} \text{ cm} = 6.58 \times 10^{-25} \text{ s}, \quad (2.3)$$

together with

$$1 \text{ GeV} = 1.78 \times 10^{-24} \text{ g} = 1.16 \times 10^{13} \text{ K} \quad (2.4)$$

for masses and temperatures. For astrophysical applications, it is useful to remember

$$M_\odot = 1.99 \times 10^{33} \text{ g} = 1.12 \times 10^{57} \text{ GeV}, \quad 1 \text{ pc} = 3.09 \times 10^{18} \text{ cm} = (6.67 \times 10^{-24} \text{ eV})^{-1}, \quad (2.5)$$

and $1 \text{ yr} = 3.16 \times 10^7 \text{ s} = (2.09 \times 10^{-23} \text{ eV})^{-1}$. In natural units, Newton's constant has dimensions of inverse energy squared and defines the Planck mass,

$$G = \frac{1}{m_{\text{pl}}^2}, \quad m_{\text{pl}} = 1.22 \times 10^{19} \text{ GeV} = 2.18 \times 10^{-5} \text{ g}. \quad (2.6)$$

(Cosmologists often use the reduced Planck mass, $M_{\text{pl}} = 1/\sqrt{8\pi G} = 2.44 \times 10^{18} \text{ GeV}$, instead.) Finally, with $\epsilon_0 = \mu_0 = 1$ (the Heaviside-Lorentz convention), the elementary charge is dimensionless, $e = \sqrt{4\pi\alpha} = 0.303$. A more complete set of conversions between natural units, SI units (kg, m, s, etc.), and astronomical units ($M_\odot$, pc, year, etc.) can be found at <https://github.com/ymzhong/naturalunit>.

The power of natural units lies in estimation: dimensional analysis plus a handful of energy scales often gives the correct answer up to $\mathcal{O}(1)$ factors. The following example is a classic.

Question 2.1. Estimate the maximum mass of a star in natural units.

Answer. Consider a star made of N nucleons with mass $m_p \simeq 1 \text{ GeV}$, supported against gravity by the degeneracy pressure of its fermions. For a star of radius R , the number density is $n \sim N/R^3$, so the Fermi momentum is $p_F \sim n^{1/3} \sim N^{1/3}/R$. When the fermions are relativistic, the kinetic energy per fermion is

$$\epsilon_{\text{kin}} \sim p_F \sim \frac{N^{1/3}}{R}, \quad (2.7)$$

while the gravitational binding energy per fermion is

$$\varepsilon_{\text{grav}} \sim -\frac{GNm_p^2}{R} = -\frac{Nm_p^2}{m_{\text{pl}}^2 R}. \quad (2.8)$$

Both terms scale as $1/R$. If the negative gravitational term wins, contraction lowers the total energy indefinitely and the star collapses. Stability therefore requires

$$N^{1/3} \gtrsim \frac{Nm_p^2}{m_{\text{pl}}^2} \Rightarrow N \lesssim \left(\frac{m_{\text{pl}}}{m_p}\right)^3 \simeq 2 \times 10^{57}. \quad (2.9)$$

The maximum stellar mass is then

$$M_{\text{max}} \sim Nm_p \sim \frac{m_{\text{pl}}^3}{m_p^2} \simeq 2 \times 10^{57} \text{ GeV} \simeq 2 M_{\odot}, \quad (2.10)$$

remarkably close to the Chandrasekhar mass, $1.4 M_{\odot}$, and obtained without a single factor of $\hbar$ or c . The mass scale of stars is fixed by just two microscopic quantities: the Planck mass and the proton mass. This is why the solar mass is the natural unit for weighing stars.

Exercise 2.1 (white dwarf radius). For a star lighter than M_{max} , the electrons supplying the degeneracy pressure are non-relativistic, and the star settles into an equilibrium configuration: a white dwarf. Repeat the estimate above with the non-relativistic degeneracy energy $\varepsilon_{\text{kin}} \sim p_F^2/(2m_e)$ and show that minimizing the total energy per fermion gives the equilibrium radius

$$R \sim \frac{m_{\text{pl}}^2}{m_e m_p^2 N^{1/3}}. \quad (2.11)$$

Evaluate R for $N = 10^{57}$ and compare it with the radius of the Earth.

2.2 Cosmology for students in a hurry

The evolution of the homogeneous and isotropic contents of the universe follows the Friedmann equation (assuming zero curvature) and the continuity equation. They are respectively given by

$$H^2 = \frac{8\pi G}{3}\rho = H_0^2 (\Omega_{\text{r},0}a^{-4} + \Omega_{\text{m},0}a^{-3} + \Omega_{\Lambda,0}), \quad \dot{\rho} = -3H(\rho + P). \quad (2.12)$$

The first equation is derived from 00 component of Einstein's equation under the Friedmann-Lemaître-Robertson-Walker metric, $ds^2 = -dt^2 + a(t)^2 \delta_{ij} dx^i dx^j$, where $a(t)$ is the scale factor. The second equation is derived from the covariant conservation of the background energy-momentum tensor $\nabla_{\mu} T^{\mu\nu} = 0$ where $T^{\mu}_{\nu} = \text{diag}(-\rho, P, P, P)$.

There is an easier way to derive the Friedmann equation that does not require general relativity. Consider a self-gravitating sphere of objects, say galaxies, with radius R and

mean density $\bar{\rho}$, and focus on its outermost shell, which has mass ΔM and expands with velocity $\dot{R}$. The kinetic energy of the shell is

$$K.E. = \frac{1}{2} \Delta M (\dot{R})^2, \quad (2.13)$$

and its gravitational potential energy is

$$P.E. = - \frac{G \left(\frac{4}{3} \pi R^3 \bar{\rho} \right) \Delta M}{R}. \quad (2.14)$$

Now suppose the outermost shell expands at exactly the escape velocity, so that its galaxies can just barely reach infinity. The total energy of the shell then vanishes,

$$0 = \frac{1}{2} \Delta M (\dot{R})^2 - \frac{G \left(\frac{4}{3} \pi R^3 \bar{\rho} \right) \Delta M}{R} \Rightarrow \left(\frac{\dot{R}}{R} \right)^2 = \frac{8\pi G \bar{\rho}}{3}. \quad (2.15)$$

By Hubble's law, the ratio $\dot{R}/R$ is the same for every shell and equals the Hubble parameter H , so we recover the first equation of Eq. (2.12). Note that the escape-velocity condition, which sets the total energy of the shell to zero, is the Newtonian counterpart of the zero-curvature assumption.

During different content-dominated eras, the universe expands differently. It evolves from an inflation era to a radiation era, then to a matter era, and finally to a dark energy era. Most of dark matter physics occurs during the radiation and matter eras. The relations between the energy density ρ , the scale factor a , the Hubble parameter H , and the cosmic time t are summarized in Tab. 1.

Content	$w \equiv P/\rho$	ρ	a	H
Radiation	1/3	$\frac{3}{32\pi G t^2} \propto a^{-4}$	$\propto t^{1/2}$	$\frac{1}{2t} \propto a^{-2}$
Matter	0	$\frac{1}{6\pi G t^2} \propto a^{-3}$	$\propto t^{2/3}$	$\frac{2}{3t} \propto a^{-3/2}$
Vacuum	-1	$\frac{3H_0^2}{8\pi G} (\text{const.})$	$e^{H_0 t}$	H_0

Table 1. Different contents of the universe, their equations of state $w \equiv P/\rho$ (pressure over energy density), and the relations between the energy density ρ , scale factor a , Hubble parameter $H \equiv \dot{a}/a$, and time t .

Question 2.2. Why does radiation have $w = 1/3$?

Answer. The photon momentum is given by $p = h/\lambda = E/c$. Radiation pressure arises from the transfer of photon momentum to an opaque boundary surface. Consider N photons with energy E moving in 1D toward the surface. The pressure is given by

$$P = \frac{F}{A} = \frac{N(p - 0)}{A\Delta t} = \frac{(nAc\Delta t)E}{A\Delta tc} = nE = \rho. \quad (2.16)$$

For 3D, we average over the incident angle θ of a photon flow on the surface,

$$P = \frac{1}{4\pi} \int d\phi \int d\cos\theta \frac{N_z p_z}{A\Delta t} = \frac{1}{2} \int_{-1}^{+1} d\cos\theta \frac{(nc\Delta t A \cos\theta) p \cos\theta}{A\Delta t} = \frac{nE}{3} = \frac{\rho}{3}. \quad (2.17)$$

Question 2.3. How does one obtain $\rho(a)$ from w ?

Answer. From the second equation of Eq. (2.12) with $P = w\rho$ and $H = \dot{a}/a$, we have

$$\frac{d\rho}{\rho} = -3(1+w)\frac{da}{a}. \quad (2.18)$$

Hence

$$\rho \propto a^{-3(1+w)}. \quad (2.19)$$

Exercise 2.2 (kination). Kination has $w = P/\rho = 1$. Derive $\rho(a)$ for kination, as well as $a(t)$, $H(t)$, and $H(a)$ for a kination-dominated universe. What is the speed of sound for a kination fluid?

From the Planck measurement of Ω_m and COBE's measurement of the CMB temperature, one can find that matter-radiation equality occurs at $a_{\text{eq}} = 2.9 \times 10^{-4}$, which corresponds to $z_{\text{eq}} = 3400$ or $t_{\text{eq}} \sim 50,000$ years after the Big Bang. This is “slightly” before recombination at $z_{\text{rec}} = 1270$ or $t_{\text{rec}} \sim 380,000$ years after the Big Bang.

Exercise 2.3. When does matter-dark energy equality occur?

3 First lecture

3.1 Evidence for dark matter

Over the past nine decades, since Zwicky first proposed dark matter to explain the unexpected concentration of galaxies in the Coma cluster [8], evidence supporting its existence has accumulated across multiple cosmic scales. Dark matter has become an indispensable ingredient of the standard cosmological model, Λ CDM, where Λ represents the cosmological constant while CDM stands for the “cold dark matter.” Dark matter has successfully explained various astronomical and cosmological observations, including the behavior of galaxy clusters, galaxy rotation curves, the Bullet Cluster collision, cosmic shear, the formation of large-scale structure (LSS), and the patterns in the cosmic microwave background (CMB). See e.g. [9, 10] for a review. Fig. 1 presents a timeline of significant events in the development of dark matter physics from 1933 to today. In this timeline, events highlighted in color represent primary observational evidence supporting the existence of dark matter. The black text shows important proposed dark matter candidates, related phenomena, or key methodologies used in dark matter research.

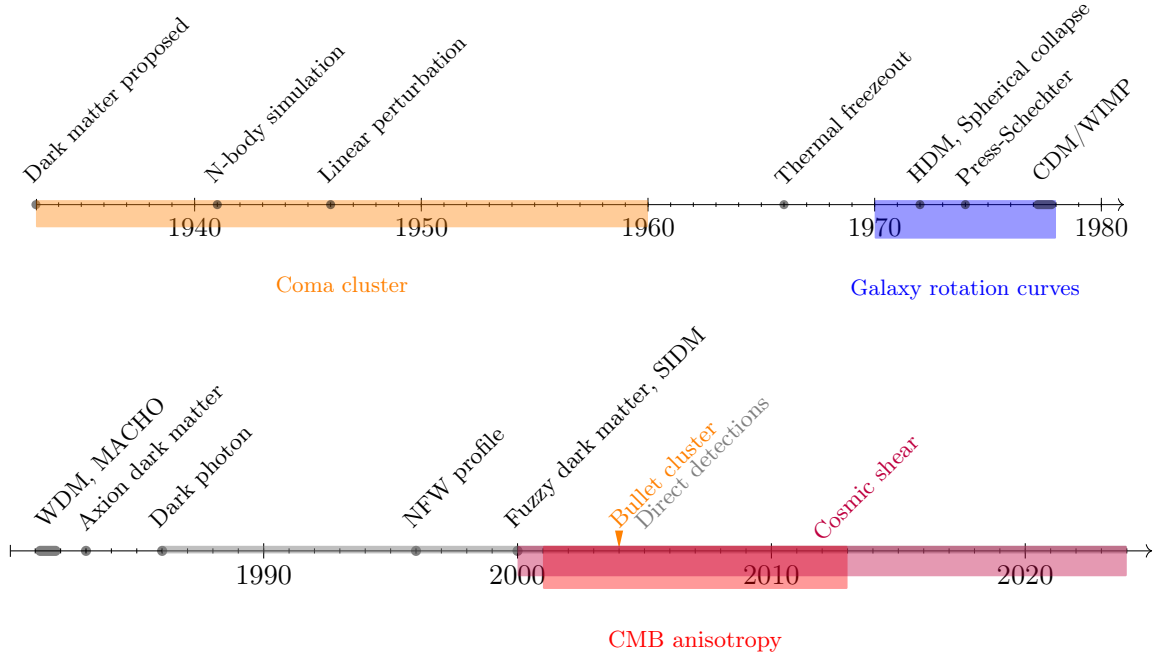


Figure 1. A brief history of dark matter and related theories and methodologies. Source: [1, 9, 10]. Acronym: CDM: cold dark matter; HDM: hot dark matter; WDM: warm dark matter; SIDM: self-interacting dark matter; WIMP: weakly-interacting massive particle; MACHO: massive compact halo object.

3.2 Basic properties of dark matter

Nevertheless, the nature of dark matter remains elusive. The statement does not mean we do not know any properties of dark matter. We have a good understanding of dark matter's behavior on large cosmological scales. To agree with observations from the CMB, Big Bang nucleosynthesis (BBN), and LSS, dark matter must be:

- Cold: dark matter is non-relativistic ($v \ll c$) during the gravitational collapse.
- Non-baryonic
- Stable
- The abundance of dark matter today is $\Omega_\chi h^2 = 0.1200 \pm 0.001$ where $h = 0.674 \pm 0.005$ is the reduced Hubble constant from Planck 2018 [11]. The abundance corresponds to a background dark matter density of $\rho_\chi = 1.3 \text{ keV cm}^{-3} \simeq 10^{-11} \text{ eV}^4$.

The abundance of dark matter is about five times that of baryonic matter. While this factor of five may seem significant, it is relatively small compared to the ratio of the dark matter abundance to the photon abundance today ($\Omega_\gamma h^2 = 2 \times 10^{-5}$), which is about 5,000. The closeness in abundance between dark matter and baryons leads to ideas such as asymmetric dark matter.

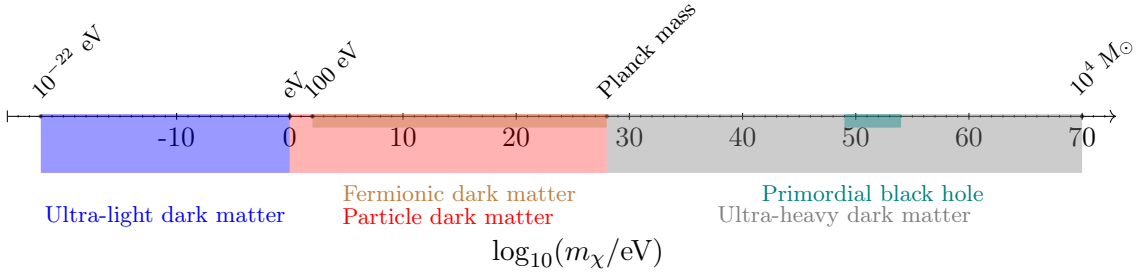


Figure 2. Important mass scales for dark matter candidates between 10^{-22} eV and $10^4 M_\odot$. We split them into ultra-light dark matter ($\lesssim 1$ eV), particle dark matter ($1 \text{ eV} - m_{\text{pl}}$), ultra-heavy dark matter ($\gtrsim m_{\text{pl}}$). We also show the mass range for fermionic particle dark matter and the primordial black hole that can constitute 100% of the dark matter.

- Adiabatic: Dark matter has the same primordial density fluctuations as baryons, photons, and neutrinos on the cosmic scale. The constraint from CMB anisotropy on the departure from adiabatic fluctuations is about $\mathcal{O}(10^{-2})$.

The adiabatic property is usually used to constrain multi-field inflation models. Some recent attempts used this property to constrain dark matter production via freeze-in [12], and later studies found the observational constraints are not strong at all [13–16].

Small-scale observations also provide estimates of dark matter’s local density and constrain dark matter’s “particle” mass, whether as an individual particle or a single composite object:

- The local dark matter density is $\rho_\chi = 0.4 \text{ GeV cm}^{-3} \simeq (0.04 \text{ eV})^4$ [17].
- Dark matter cannot be too light, $m \gtrsim 10^{-22} \text{ eV}$, as its de Broglie wavelength would then exceed the virial radius of a dwarf galaxy. Otherwise, the bulk of the dwarf halo will reside in the solitonic region with an ultra-high central density and an ultra-small half-mass radius, which is in tension with observations [7].
- Dark matter cannot be too heavy, $m \lesssim 10^4 M_\odot$, as there should be a sufficient number of particles ($N \gg 10^6$) to populate a dwarf galaxy (for a typical mass of $10^{10} M_\odot$) such that its density distribution and gravitational field are mostly smooth and do not show granularity. If the number of dark matter particles $N \ll 10^6$, the granularity of the gravitational interactions shows up and consequent perturbations exerting on a star will lead to a total loss of memory of the initial condition of the stellar orbit, within the life time of a dwarf galaxy. In such a scenario, we will observe dwarf stars always being spherical shaped (globular-cluster-like).¹

The mass of dark matter “particles” spans over 90 orders of magnitude ($1 M_\odot \simeq 10^{66} \text{ eV} \simeq 2 \times 10^{33} \text{ g}$). However, it is still finite. Fig. 2 shows the landscape of dark matter mass. Note that there are a few important mass scales in this vast range:

¹The same argument leads to the spherical-shaped core of a self-interacting dark matter halo, except there the perturbations are driven by interactions between dark matter particles.

- $m_\chi = m_{\text{pl}} \simeq 10^{28} \text{ eV} \simeq 22 \mu\text{g}$, above which a single dark matter particle’s Compton wavelength will be within its Schwarzschild radius

$$m_\chi \gtrsim m_{\text{pl}} \Rightarrow \frac{1}{m_\chi} \lesssim Gm_\chi = \frac{m_\chi}{m_{\text{pl}}^2}. \quad (3.1)$$

Therefore, for a dark matter “particle” with $m_\chi > m_{\text{pl}}$, what we mean by “particle” is not a particle but a single composite object, such as a primordial black hole or a bag of particles. We call those candidates ultra-heavy dark matter.

- $m_\chi = 100 \text{ eV}$. If dark matter is fermionic, a halo can be considered a self-gravitating degenerate Fermi gas system. (Derivation is similar to Question 2.1) For such a halo to exist, the Fermi velocity v_F , which is given by

$$v_F = \sqrt{\frac{2E_F}{m_\chi}} = \frac{1}{m_\chi} \left(\frac{6\pi^2 n_\chi}{g_\chi} \right)^{1/3} \quad (3.2)$$

where $E_F = \frac{1}{2m_\chi} \left(\frac{6\pi^2 n_\chi}{g_\chi} \right)^{2/3}$ is the Fermi energy and g_χ is the number of spin and flavor states, should not exceed the escape velocity of the halo. Assuming a halo has a uniform density ($\rho_\chi = n_\chi m_\chi$ is fixed), the condition yields

$$v_F \leq \sqrt{\frac{2GM_h}{r_h}} \Rightarrow m_\chi \gtrsim g_\chi^{-1/4} G^{-3/8} M_h^{-1/8} r_h^{-3/8}, \quad (3.3)$$

where M_h and r_h are the halo mass and radius, respectively. Eq. (3.3) indicates that stronger constraints can be obtained from a smaller halo. Ref. [18] analyzes Leo II, a dwarf galaxy, and imposes the constraint $m_\chi \geq 0.13 \text{ keV}$ for fermionic dark matter with $g_\chi = 2$.

- $m_\chi = 1 \text{ eV}$, below which the particles’ de Broglie wavelengths overlap.

$$\frac{1}{m_\chi v_\chi} \geq \left(\frac{\rho_\chi}{m_\chi} \right)^{-1/3} \Rightarrow m_\chi \leq \rho_\chi^{1/4} v_\chi^{-3/4} \sim \mathcal{O}(\text{eV}). \quad (3.4)$$

Sub-eV dark matter particles should be thought of as wave-like (like classical electromagnetic field) rather than particle-like. Those candidates are known as ultra-light dark matter.

What we mean by “the nature of dark matter remains elusive” is that we do not understand the properties of dark matter as comprehensively as those of the Standard Model particles. Tab. 2 compares dark matter to neutrinos, the least understood corner of ordinary matter. Unlike neutrinos, we still lack an understanding of dark matter’s spin, the number of species, and production mechanisms. Constraints on the dark matter mass and its local number density are much looser than those of neutrinos. We are also unsure if dark matter can be produced on Earth or in faraway stars.

The elusive nature of dark matter is both a challenge and an opportunity. It stimulates theorists to propose new models and enables experimentalists to invent new search schemes. Meanwhile, despite the limited understanding of dark matter, people have been using the distribution of dark matter to probe what happens in the early universe and other fundamental physics.

	Neutrinos	Dark Matter
Number of species	Three flavors	?
Mass	$0 - 0.8$ eV	$10^{-22} - 10^{70}$ eV
(Sum of masses)	$0.059 - 0.064$ eV	??
Spin	$1/2$	?
Electric charge	0	$\lesssim 10^{-10} - 10^{-1}$
Self-interaction	Electroweak & ?	Gravity & ?
Int. w/ other particles	Electroweak & ?	Gravity & ?
Lifetime	$\gtrsim 6 \times 10^{15}$ s	$\gtrsim 4.4 \times 10^{17}$ s
Decay channel	?	?
Cosmic production	Decoupled from SM plasma at $T \sim 1$ MeV	?
Other prod. channel	Stars, cosmic rays, reactors, accelerators ...	?
Abundance today	$(1.4 - 2.7) \times 10^{-3}$	0.264
Abundance at recomb.	0.1	0.76
% of the total mass	$(0.4 - 0.9)\%$	84%
ρ_{bkg}	$(6.7 - 13)$ eV cm $^{-3}$	1300 eV cm $^{-3}$
n_{bkg}	$3 \times 2 \times 56$ cm $^{-3}$	$10^{-67} - 10^{25}$ cm $^{-3}$
Clustering	No (?)	Yes
ρ_{local}	$\sim \rho_{\text{bkg}}$	0.4 GeV cm $^{-3}$
n_{local}	$\sim n_{\text{bkg}}$	$10^{-62} - 10^{30}$ cm $^{-3}$
Momentum dist.	Fermi-Dirac	?

Table 2. Properties of neutrinos and dark matter [1, 19, 20]. The neutrino column’s lower section lists the properties of cosmic relic neutrinos. We cite the limits from DESI baryon acoustic oscillation (BAO) + CMB for the total neutrino mass, assuming the neutrino masses follow the normal hierarchy [21]. The narrow allowed parameter region between the masses inferred from oscillation experiments and those derived from DESI under Λ CDM is often referred to as the “*neutrino mass problem*.” The cosmic neutrino abundance and its percentage of the total mass are calculated accordingly with cosmological parameters from [11].

3.3 Dark matter production: thermal freeze-out (a story too good to be true)

3.3.1 Thermodynamics in the early universe

Given a distribution function of a system of particles, $f(p, T)$, we can obtain its macroscopic quantities by taking moments of the distribution function, i.e., integrating over the phase space with some kernel functions. The number density, energy density, and pressure are the moments considered in many Einstein-Boltzmann equations. For a given $f(p, T)$, they are respectively given by

$$n(T) = \frac{g}{(2\pi)^3} \int d^3p f(p, T), \quad (3.5)$$

$$\rho(T) = \frac{g}{(2\pi)^3} \int d^3p f(p, T) E(p), \quad (3.6)$$

$$P(T) = \frac{g}{(2\pi)^3} \int d^3p f(p, T) \frac{p^2}{3E(p)}. \quad (3.7)$$

For bosons and fermions in the early universe, their resulting number density, energy density, and pressure, under the relativistic limit, are respectively given by

$$n = \frac{\zeta(3)}{\pi^2} g T^3 \begin{cases} 1 & \text{boson} \\ \frac{3}{4} & \text{fermion} \end{cases}, \quad \rho = \frac{\pi^2}{30} g T^4 \begin{cases} 1 & \text{boson} \\ \frac{7}{8} & \text{fermion} \end{cases}, \quad P = \frac{\rho}{3} \quad (3.8)$$

If they are non-relativistic, these quantities change to

$$n = g \left(\frac{mT}{2\pi} \right)^{3/2} e^{-m/T}, \quad \rho = mn + \frac{3}{2} nT, \quad P = nT, \quad (3.9)$$

which are all Boltzmann suppressed ($\sim e^{-m/T}$).

If there are multiple relativistic species, the energy density changes to

$$\rho = \frac{\pi^2}{30} g_*(T) T^4 \quad (3.10)$$

where

$$g_* = \sum_{i=b} g_i \left(\frac{T_i}{T} \right)^4 + \frac{7}{8} \sum_{i=f} g_i \left(\frac{T_i}{T} \right)^4 \quad (3.11)$$

The Standard Model at high temperature has 28 degrees of freedom for bosons and 90 degrees of freedom for fermions. Hence the corresponding $g_* = 106.75$.

From the first law of thermodynamics, we have $TdS = dU + PdV$. Now defining the entropy density (or specific entropy) $s \equiv S/V$ and substituting it together with $U = \rho V$ into the first law gives

$$(Ts - \rho - P)dV + V \left(\frac{Tds}{dT} - \frac{d\rho}{dT} \right) dT = 0 \quad (3.12)$$

For the equation to always hold, we need

$$s = \frac{\rho + P}{T}, \quad \frac{d(sa^3)}{dt} = 0, \quad (3.13)$$

where the derivation of the second equation uses the continuity equation. The first equation tells us how to compute the specific entropy and the second equation demands that entropy be conserved.

To compute the specific entropy for multiple relativistic species, from Eq. (3.10), we have

$$s = \frac{2\pi^2}{45} g_{*S}(T) T^3, \quad (3.14)$$

where

$$g_{*S} = \sum_{i=b} g_i \left(\frac{T_i}{T} \right)^3 + \frac{7}{8} \sum_{i=f} g_i \left(\frac{T_i}{T} \right)^3. \quad (3.15)$$

In such a scenario, entropy conservation, $d(sa^3)/dt = 0$, implies

$$g_{*S}(T) T^3 a^3 = \text{const}, \quad \text{or} \quad T \propto g_{*S}^{-1/3} a^{-1}. \quad (3.16)$$

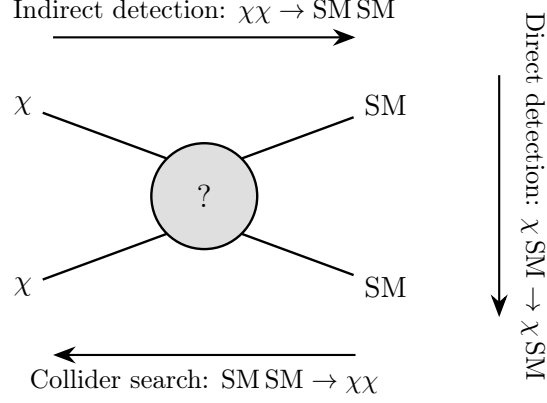


Figure 3. Schematic summary of the complementary approaches used in searches for WIMP dark matter. Reading the same interaction along different time directions gives indirect detection (rightward), collider search (leftward), and direct detection (downward).

It will be convenient to define the abundance $Y_i \equiv n_i/s$, which removes the a -dependence if there is no particle creation or annihilation. If dark matter is produced thermally, its abundance Y_χ should not change from its freeze-out to today. Given $\rho_{\chi,0} = 10^{-11} \text{ eV}^4 = m_\chi n_{\chi,0}$ and $s_0 \simeq 10^{-11} \text{ eV}^3$, we have

$$Y_\chi \equiv \left(\frac{n_\chi}{s} \right)_0 = \frac{\text{eV}}{m_\chi}. \quad (3.17)$$

3.3.2 Dark matter thermal freeze-out and the WIMP miracle

The dark matter thermal freeze-out happens when the Hubble parameter catches up with the annihilation rate of dark matter particles into the Standard Model particles.

$$n_\chi \langle \sigma v \rangle \sim H \sim \frac{T_{\text{fo}}^2}{m_{\text{pl}}}, \quad (3.18)$$

where T_{fo} is the freeze-out temperature, which is around $m_\chi/10$. From Eq. (3.18), one can estimate the dark matter abundance Y_χ at freeze-out,

$$(Y_\chi)_{\text{fo}} = \frac{n_{\chi,\text{fo}}}{s_{\text{fo}}} \sim \frac{T_{\text{fo}}^2}{m_{\text{pl}} \langle \sigma v \rangle} \frac{1}{T_{\text{fo}}^3}. \quad (3.19)$$

From Eq. (3.17), we also have $(Y_\chi)_{\text{fo}} = Y_\chi = \frac{\text{eV}}{m_\chi}$. Combining these two equations and $T_{\text{fo}} \sim m_\chi/10$, we have

$$\langle \sigma v \rangle \sim \frac{10}{\text{eV} m_{\text{pl}}}. \quad (3.20)$$

More careful computation shows that the annihilation cross section needed to obtain the correct dark matter relic abundance is given by

$$\langle \sigma v \rangle = 3 \times 10^{-26} \text{ cm}^3/\text{s} = \frac{1}{(20 \text{ TeV})^2}. \quad (3.21)$$

The cross section seems to indicate dark matter physics should be at the electroweak scale. Such dark matter candidates are so-called weakly interacting massive particles (WIMPs).

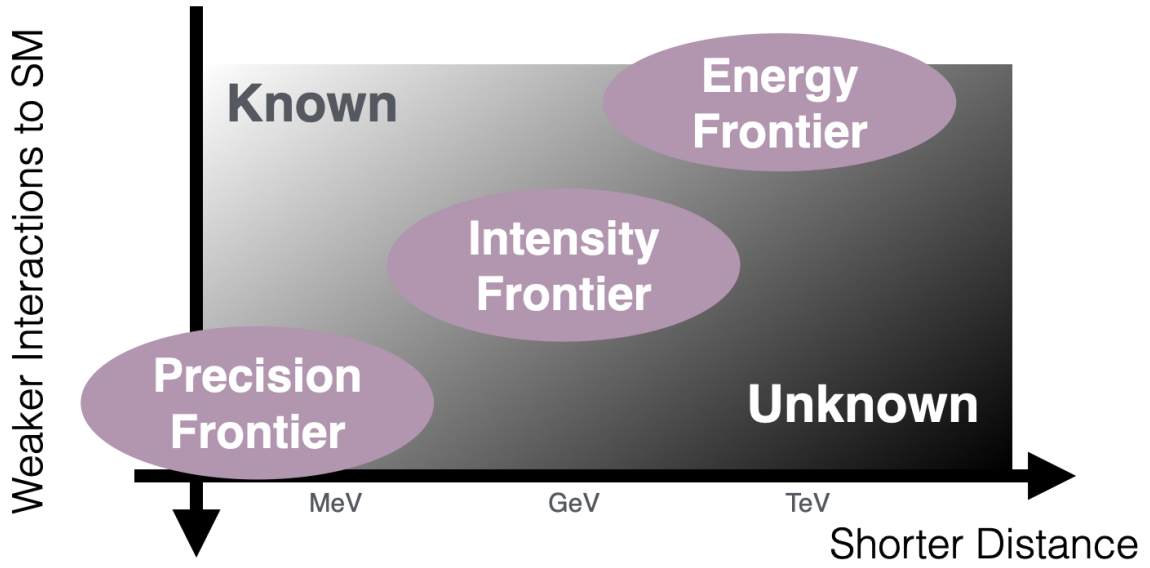


Figure 4. Frontiers in particle physics

WIMPs are predicted by supersymmetry models, a class of theories that also address the electroweak hierarchy problem and grand unification.

More importantly, the production mechanism also indicates complementary approaches to detect WIMP dark matter. These include direct detection, indirect detection, and collider searches. However, three decades of WIMP searches have yielded no conclusive evidence. It is time to go beyond the WIMP paradigm.

3.4 Exodus: Dark matter with dark sectors

The Standard Model includes numerous bosons and fermions with intricate interactions. A similar situation might occur with dark matter, potentially existing within dark sectors composed of feebly interacting particles, such as dark photons and dark Higgs. The concept of dark sectors can be traced back to the hidden sectors involved in supersymmetry breaking (see review [22]) and the hidden valleys [23]. This idea gained popularity when [24] employed the Sommerfeld-enhanced dark matter annihilation, a phenomenon involving a light dark mediator, to explain the PAMELA positron excess [25]. Over the past decade, light dark sector particles have emerged as crucial ingredients in addressing numerous other puzzles in particle physics, such as the muon $g-2$, neutron lifetime [26], proton charge radius [27], and the X17 ATOMKI anomaly [28]. Among the notable examples of the dark sector particles are the “*fabulous five*”:²

- Dark photon, A' ,
- Dark Higgs, S ,
- Axion-like particle, a ,

²Source: Prof. Lian-Tao Wang.

- Sterile neutrino, N ,
- Millicharged particle.³

Their interactions with the Standard Model sector are through various “portals”:

- Vector portal with kinetic mixing: $\epsilon F_{\mu\nu} F'^{\mu\nu}$
- Vector portal with $U(1)_{B-L}$: $A'_\mu J_{B-L}^\mu$
- Higgs portal: $H^\dagger H(\lambda S^2 + AS)$
- Neutrino portal: LHN
- Axion portal: $\frac{1}{f}\partial_\mu a J_A^\mu$
- ...

The growing interest in dark sector particles and portal interactions signifies a paradigm shift in the particle physics community. This new perspective acknowledges that new physics can occur at both higher and lower energies (see Fig. 4). Consequently, we should fully leverage available facilities and technologies to explore new light physics. This shift is evident in the blooming of intensity frontier experiments, such as flavor factories, fixed-target experiments, muon experiments, and neutrino experiments, as well as precision frontier experiments, including the muon $g - 2$, electron’s electric dipole moment, and the fifth force search.

3.5 Dark sector complementarity

Let us consider a concrete example to illustrate the complementarity of different probes to search for a light dark sector. We take the dark matter candidate to be a Dirac fermion χ , with a *real* scalar mediator S that couples only to Standard Model muons.

$$\mathcal{L} \supset -g_\chi S \bar{\chi}\chi - g_\mu S \bar{\mu}\mu. \quad (3.22)$$

From a UV point of view, for new physics scales $\Lambda \gtrsim 1$ TeV, the muon couplings $g_\mu \lesssim \mathcal{O}(10^{-4} - 10^{-3})$. The S -muon couplings in (3.22) also introduce effective scalar couplings to a pair of vector bosons (SVV) at one-loop. For a light scalar with mass below the electroweak scale, the most relevant SVV coupling is the scalar-diphoton coupling, $S\gamma\gamma$. See [29] for more details.

Let us turn our attention to dark matter. There are three distinct possibilities for the relative sizes of m_χ and m_S , each carrying different phenomenological consequences:

- $m_\chi < m_S/2$: In this case, if $g_\chi \gtrsim g_\mu$, the mediator S will primarily decay invisibly to $\chi\bar{\chi}$. The thermal freeze-out relic abundance of dark matter is driven by s -channel

³The millicharged particle can be considered a new particle coupled to a massless dark photon.

annihilation into leptons, $\bar{\chi}\chi \rightarrow \mu^+\mu^-$ (or $\gamma\gamma$) in the early Universe. The annihilation rate is roughly given by

$$\langle\sigma v\rangle = 2\pi \frac{\alpha_D \alpha_\mu}{m_\chi} \frac{(m_\chi^2 - m_\mu^2)^{3/2}}{(m_S^2 - 4m_\chi^2)^2} \langle v_{\text{rel}}^2 \rangle, \quad (3.23)$$

where $\alpha_{D(\mu)} \equiv g_{D(\mu)}^2/(4\pi)$ and $\langle v_{\text{rel}}^2 \rangle$ is the thermal average of the relative dark matter velocity squared, reflecting the fact that annihilation is a p -wave process for a scalar mediator with a Dirac fermion dark matter.

Question 3.1. Why do we prefer p -wave annihilation?

Answer. In the non-relativistic limit, the annihilation cross-section can be expanded in powers of the relative velocity,

$$\sigma v_{\text{rel}} = \underbrace{a}_{s\text{-wave}} + \underbrace{b v_{\text{rel}}^2}_{p\text{-wave}} + \mathcal{O}(v_{\text{rel}}^4), \quad (3.24)$$

where the velocity-independent term a corresponds to s -wave annihilation and the $b v_{\text{rel}}^2$ term to p -wave annihilation. Taking the thermal average over the dark matter velocity distribution, with $\langle v_{\text{rel}}^2 \rangle = 6T/m_\chi$, gives

$$\langle\sigma v\rangle = a + \frac{6b}{x} + \mathcal{O}(x^{-2}), \quad x \equiv \frac{m_\chi}{T}. \quad (3.25)$$

At freeze-out, $x_{\text{fo}} \simeq 20$, so the p -wave term is only mildly suppressed, $\langle v_{\text{rel}}^2 \rangle \simeq 0.3$. Observations of the CMB constrain the amount of energy injected into the Intergalactic Medium (IGM) through dark matter annihilation at late times, which can distort the CMB. By the recombination era, however, the typical dark matter velocity has redshifted to a value many orders of magnitude below its freeze-out value (after kinetic decoupling, $v_\chi \propto a^{-1}$). An s -wave cross-section, $\langle\sigma v\rangle = a$, is velocity independent and thus strongly constrained [30–32], while a p -wave cross-section is suppressed by v_{rel}^2 and can effectively evade these CMB constraints. This is why the $\langle v_{\text{rel}}^2 \rangle$ factor appears in Eq. (3.23): a scalar mediator with Dirac fermion dark matter annihilates in the p -wave.

- $m_S/2 \lesssim m_\chi \lesssim m_S$: In this case, s -channel annihilation into leptons again sets the relic abundance of χ , providing a thermal relic target. However, $S \rightarrow \chi\bar{\chi}$ decays are kinematically forbidden, so S will decay visibly.
- $m_S < m_\chi$: In this case, S again decays visibly. However, annihilation in the early Universe will primarily proceed through secluded annihilation, $\bar{\chi}\chi \rightarrow SS$. The cross-section for this process only depends on the dark sector coupling g_χ , so there is no well-defined thermal relic target for terrestrial experiments.

Here we focus on the first scenario with $m_\chi < m_S/2$, since it provides concrete thermal targets and is generally difficult to test, given the invisible decays of S . Fig. 5 from Ref. [29]

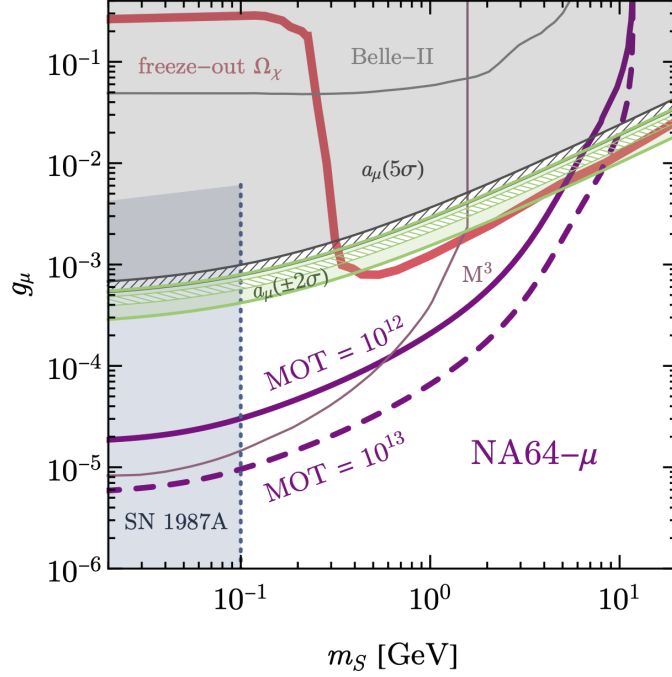


Figure 5. Current (shaded regions) and projected (lines) constraints for the dark matter-muon-specific mediator model in the $g_\mu - m_S$ parameter space. We assume $m_\chi = m_S/3$ and $g_\chi = 1$. All the constraints are at 95% CL. The red curve indicates the parameters required to reach the correct thermal freeze-out dark matter abundance. The green and gray regions represent the 2σ -favored and 5σ -excluded regions based on current a_μ measurements. The blue region represents approximate exclusions from the cooling of SN 1987A. Colored lines are the sensitivity projections for NA64- μ , M^3 [33], and mono-photon searches at Belle-II [34].

presents various constraints on the dark sector model assuming $m_\chi = m_S/3$ and $g_\chi = 1$. It is a busy plot. The various shaded regions and lines indicate existing and future probes of the model, respectively, forming a tight complementarity for the parameter space. We will summarize the major constraints as the “floor”, the “ceiling”, the “fence”, and the “walls” at the lighter end of the dark scalar/dark matter mass.

- **The floor: Relic abundance.** Regions below the red curves are excluded due to an overabundance of dark matter. This can be understood as follows: for smaller couplings, the annihilation rate is more easily overtaken by the universe’s expansion, ending the freeze-out at a higher temperature. For a rough estimate, set

$$\langle\sigma v\rangle \sim 2\pi\alpha_D\alpha_\mu\frac{m_\chi^2}{m_S^2} \sim 3 \times 10^{-26} \text{ cm}^3/\text{s} = \frac{1}{(20 \text{ TeV})^2} \quad (3.26)$$

yielding $g_\mu \sim 10^{-3}$ at $m_S = 1 \text{ GeV}$ ($m_S/m_\chi = 3$, $\alpha_D = 1/4\pi$). The benchmark value $\langle\sigma v\rangle$ is derived from the observed dark matter relic abundance under the dark matter freeze-out scenario. See Sec. 3.3.2 for more details. The kink in the red curve around $m_S = 2m_\mu$ occurs when a new annihilation channel ($\chi\chi \leftrightarrow \mu^+\mu^-$) becomes kinematically accessible in addition to $\chi\chi \leftrightarrow \gamma\gamma$, which allows a drop in the couplings.

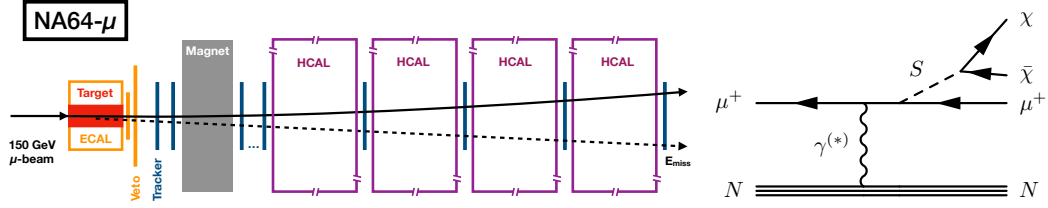


Figure 6. Left: The detector segment of the proposed NA64- μ experimental setup. Right: A diagram that leads to a muon missing momentum event.

Note that the constraints from the relic abundance could change dramatically due to the resonant enhancement, which occurs when $m_S \approx 2m_\chi$. In such a limit, $\langle\sigma v\rangle$ can be greatly enhanced, and the overabundance floor will move downward considerably. There are many other tricks (such as coannihilation) to alter the relic abundance of thermal dark matter [35].

- **The ceiling: Searches in collider and accelerator, rare decays.** The best way to search for light dark sector particles is through intensity frontier experiments, including fixed target experiments, rare decays, flavor factories, high-luminosity collider experiments, and neutrino experiments. Because we assume dark scalars predominantly couple to dark matter rather than Standard Model particles (whose couplings are severely constrained), the signals we look for include missing momentum/energy, partially invisible decays, or mono-photon signals.

Take NA64- μ , the fixed-target experiment with a muon beam at CERN, as an example. The setup is shown in the left panel of Fig. 6. It shoots a 100 GeV muon beam at a lead target and looks for events with large missing momentum using the trackers and hadronic calorimeter behind the target. If the dark scalar is produced through the bremsstrahlung process $\mu N \rightarrow \mu N S(\bar{\chi}\chi)$ (right panel of Fig. 6), it will appear as a missing momentum event. For $N_\mu = 10^{12}$ total number of muons on target, requiring at least three missing momentum events to be observed leads to $g_\mu \sim 10^{-5}$. For a heavy mediator, the beam does not have enough energy to produce it, so NA64- μ loses sensitivity to scalars with mass $\gtrsim \mathcal{O}(1)$ GeV.

Exercise 3.1. Estimate g_χ given the setup of NA64- μ . *hint:* The cross section $\mu N \rightarrow \mu N S(\bar{\chi}\chi)$ can be treated using the improved Weizsacker-Williams approximation, where the virtual photon between μ and N can be treated as a real photon and the $2 \rightarrow 3$ process can be factorized.

- **The left fence: SN 1987A.** A crucial constraint on light scalars coupled to the Standard Model particles arises from supernova (SN) cooling. A core-collapse supernova hosts a proto-neutron star, with highly degenerate and relativistic electrons, near-degenerate and non-relativistic nucleons, and muons that are not so degenerate [36]. It primarily cools through neutrino diffusion. The measured SN 1987A

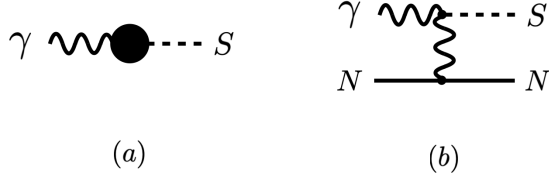


Figure 7. Processes that generate dark scalars in the SN1987A. (a) resonant mixing; Note that photons in the plasma carry an in-medium mass. Its longitudinal mode can mix with the scalar. (b) Primakoff process.

neutrino burst flux matches model predictions [37]. The energy loss per unit mass is constrained by the so-called “Raffelt bound”,

$$\epsilon \lesssim \epsilon_{\text{Raffelt}} = 10^{19} \text{ erg g}^{-1} \text{ s}^{-1}. \quad (3.27)$$

The SN event can constrain S and χ produced through prompt decays. If $g_\mu(g_\gamma)$ is too small, no significant dark scalar population is produced in the SN, so S does not contribute to cooling. If $g_\mu(g_\gamma)$ is too large, dark matter produced via S decays is trapped inside the SN and again does not contribute to cooling.

Dark scalars with mass less than the plasma frequency of the photon, ω_p ($\simeq 20$ MeV for SN 1987A), can be resonantly produced through mixing with the in-medium photon’s longitudinal mode [38]. See left panel of Fig. 7. Requiring the energy loss to be below this value gives an upper limit in the $m_S - g_\mu$ plane (lower edge of the excluded region) around $g_\mu \approx 10^{-8}$ for $m_S \lesssim 20$ MeV [39]. For $m_S \sim 100$ MeV, resonant production is suppressed, but dark scalars can still be produced via continuum processes like $\gamma N \rightarrow NS$ [40]. See right panel of Fig. 7. This results in an upper limit around $g_\mu \approx \text{few} \times 10^{-7}$ for $m_S \lesssim 100$ MeV. Dark scalars with mass above 100 MeV are too heavy to be produced in a supernova.

Once produced, dark scalars decay into $\chi\bar{\chi}$, given $m_S > 2m_\chi$ and $g_\mu \ll g_\chi = 1$. Dark matter interacts with supernova plasma particles, such as muons, photons, and protons, limiting dark matter outflow and setting a lower limit (upper edge of the excluded region) on g_μ for sufficiently light scalars produced in the SN. For a rough estimate, we only consider $\chi\gamma \rightarrow \chi\gamma$ interactions and adopt a mean-free-path criterion,

$$(n_\gamma \sigma_{\chi\gamma \rightarrow \chi\gamma})^{-1} \lesssim r_{\text{core}}. \quad (3.28)$$

Adopting $n_\gamma \approx 10^{33} \text{ cm}^{-3}$ and $r_{\text{core}} \approx 1 \text{ km}$ yields $g_\mu \sim \mathcal{O}(10^{-3})$. A more sophisticated treatment uses the “ $\pi/2$ -deflection criterion” proposed by [41], leading to $g_\mu \simeq 10^{-4}$.

3.6 The wall: N_{eff} constraints

BBN begins when the universe’s temperature drops below $T \sim 1$ MeV and concludes around $T = 30$ keV. The critical stages of standard BBN include (I) neutrino decoupling ($t \sim 0.1$ sec), (II) n/p freeze-out ($t \sim 1$ sec), (III) e^+e^- annihilation ($t \sim 1$ sec), and (IV) the deuterium bottleneck ($t \sim 3$ min). Figure 8 shows the evolution of the abundance of

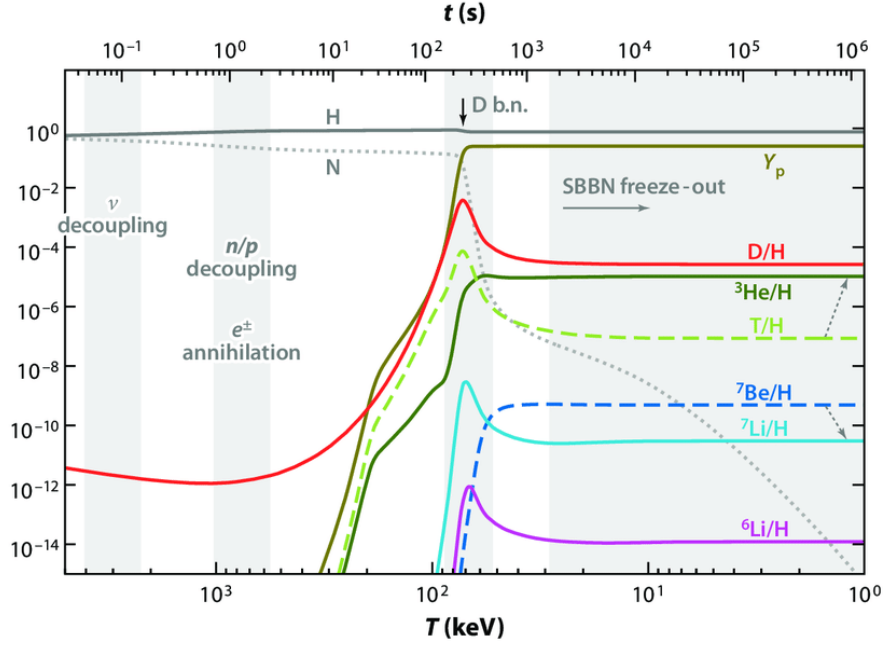


Figure 8. The evolution of the abundance of primordial elements during the standard BBN process. The gray band indicates four important stages: neutrino decoupling, n/p freeze-out and electron-positron annihilation, deuterium bottleneck, and freeze-out of all other elements. The plot is taken from [42].

primordial elements during the standard BBN process [42]. These theoretical predictions are tested against observations using two key observables: the primordial ${}^4\text{He}$ mass fraction, Y_p , and the ratio of the abundances of primordial deuterium and hydrogen, D/H . The theoretical predictions for the standard BBN are [43, 44]:

$$Y_p = 0.2469 \pm 0.0002, \quad D/H = (2.506 \pm 0.110) \times 10^{-5}. \quad (3.29)$$

The current measured values are:

$$Y_p = 0.245 \pm 0.003, \quad D/H = (2.547 \pm 0.029) \times 10^{-5}. \quad (3.30)$$

A third piece of check is the baryon to photon ratio, η . BBN yields

$$\eta = (6.040 \pm 0.118) \times 10^{-10}, \quad (3.31)$$

which is in agreement with the more precise measurement from Planck CMB $\eta = (6.12 \pm 0.04) \times 10^{-10}$ [43]. Therefore, the standard BBN model is well-supported by the observational data.⁴

⁴Recent progress in addressing the overabundance of primordial ${}^7\text{Li}$ can be found in [43, 45].

Question 3.2. Why the BBN begins at $T \sim 1$ MeV?

Answer. At high temperature, neutron and proton is in equilibrium with weak interaction. The n/p inter-conversion is given by

$$\Gamma \sim G_F^2 T^5 \quad (3.32)$$

where G_F is the Fermi constant. On the other end, the Hubble rate for the radiation dominated area is given by

$$H \sim \sqrt{g_*} T^2 / M_{\text{pl}} \quad (3.33)$$

The freezeout of neutrons and protons happens when $\Gamma < H$, which is

$$T_{\text{fr}} \sim g_*^{1/6} M_{\text{pl}}^{-1/3} G_F^{-2/3} \sim 1 \text{ MeV}. \quad (3.34)$$

A more precise calculation yields $T_{\text{fr}} = 0.8$ MeV. Therefore the ratio of neutrons to protons, at freeze-out is given by $n/p = \exp((m_p - m_n)/T_{\text{fr}}) \simeq 1 : 5$. This ratio further drops with respect to time due to neutron decays. At 3 min, it drops to $n/p \simeq 1 : 7$. At that time, all the neutrons goes to Helium, hence we should have

$$Y_p = \frac{2n}{p+n} \simeq 0.25. \quad (3.35)$$

BBN and CMB are crucial for constraining light dark matter and dark sector models, including those coupled to the Standard Model via dark mediators or pure gravity. The new physics manifests itself in a quantity, N_{eff} , that relates the energy density of photons, ρ_γ , to the total radiation energy density, ρ_{rad} :

$$\rho_{\text{rad}} = \rho_\gamma \left(1 + \frac{7}{8} \left(\frac{4}{11} \right)^{1/3} N_{\text{eff}} \right) \Rightarrow N_{\text{eff}} \equiv \frac{8}{7} \left(\frac{11}{4} \right)^{4/3} \left(\frac{\rho_{\text{rad}} - \rho_\gamma}{\rho_\gamma} \right). \quad (3.36)$$

The pre-factor $\frac{7}{8} \left(\frac{4}{11} \right)^{1/3}$ before N_{eff} is designed explicitly for neutrinos, which were in thermal equilibrium with the Standard Model plasma until the weak interaction rate became smaller than the Hubble parameter around $T = \mathcal{O}(1)$ MeV. After decoupling, the cosmic neutrino relic undergoes expansion and maintains a quasi-thermal distribution with a temperature $T_\nu \propto a^{-1}$. Photons have the same scale-factor dependence, $T_\gamma \propto a^{-1}$. Without electron-positron annihilation, both fluids would share the same temperature.

In the standard BBN, electron-positron annihilation occurs around $T \sim 0.5$ MeV. The process heats the photons and creates an interesting ratio between T_γ and T_ν . Between neutrino decoupling and electron-positron annihilation, the total entropy degrees of freedom (dof) of the Standard Model plasma is given by (see Sec. 2.2 for more details)

$$g_{*s}(T_1) = \underbrace{2}_{\gamma\text{'s dof}} + \underbrace{\frac{7}{8}}_{\text{Fermion discount}} \left(\underbrace{2}_{e^+\text{'s dof}} + \underbrace{2}_{e^-\text{'s dof}} \right) = \frac{11}{2}. \quad (3.37)$$

After electron-positron annihilation, the total entropy dof becomes

$$g_{*s}(T_2) = \underbrace{2}_{\gamma\text{'s dof}}. \quad (3.38)$$

Entropy conservation demands

$$g_{*s}(T_1)T_1^3 = g_{*s}(T_2)T_2^3 \quad \Rightarrow \quad T_\nu = \left(\frac{4}{11}\right)^{1/3} T_\gamma, \quad (3.39)$$

where ν 's follow the temperature T_1 and γ 's follow the temperature T_2 . Therefore, if there are N_{eff} species of neutrinos, their contribution to ρ_{rad} is

$$N_{\text{eff}} \times \rho_\nu = N_{\text{eff}} \times \frac{7}{8} \left(\frac{4}{11}\right)^{4/3} \times \rho_\gamma, \quad (3.40)$$

where the photon energy density and the neutrino density (one species) are respectively given by

$$\rho_\gamma = 2 \times \frac{\pi^2}{30} T_\gamma^4, \quad \rho_\nu = \frac{7}{8} \times 2 \times \frac{\pi^2}{30} T_\nu^4. \quad (3.41)$$

Hence, we recover the second term in the RHS bracket of Eq. (3.36). If neutrinos decouple instantaneously, the standard BBN yields $N_{\text{eff}} = 3$. A more precise treatment of neutrino decoupling considers non-instantaneous decoupling and the process $e^+e^- \rightarrow \nu\bar{\nu}$ during the electron-positron annihilation. This yields a slightly larger value [46]

$$N_{\text{eff}}^{\text{SM}} = 3.044 \pm 0.0005. \quad (3.42)$$

If we have extra radiation sources besides photons and neutrinos, we can define

$$\Delta N_{\text{eff}} \equiv \rho_{\text{extra rad}}/\rho_\nu \quad (3.43)$$

and simplify Eq. (3.36) into

$$N_{\text{eff}} = 3.044 + \Delta N_{\text{eff}}. \quad (3.44)$$

Observationally, N_{eff} (or ΔN_{eff}) can be constrained in two ways. By carefully studying the nuclear reaction processes during BBN for different N_{eff} and comparing them to the observations of D/H and Y_p , the limit on N_{eff} from BBN [44] is given by:

$$\Delta N_{\text{eff}}^{\text{BBN}} = 0.407. \quad (3.45)$$

Note that the value is the one-sided 95% confidence level limit.

The second method relies on the CMB anisotropy. It yields the ratio $(\rho_{\text{rad}} - \rho_\gamma)/\rho_\gamma$ by measuring the angular size of the sound horizon θ_s and the angular size of the photon diffusion length θ_d [47]. The current limit from Planck 2018 (quoted from [44]) is:

$$\Delta N_{\text{eff}}^{\text{CMB}} = 0.513. \quad (3.46)$$

Again, the value is the one-sided 95% confidence level limit.

Besides anisotropy, another important measurement of the CMB is the CMB temperature today, $T_{\gamma,0} = 2.73$ K, as measured by COBE/FIRAS [48]. This measurement fixes the energy density of the Standard Model electromagnetic sector ρ_γ . We would like to emphasize that $T_{\gamma,0}$ is the only temperature we have measured from the cosmological observations.

Now let us consider two examples where a light dark matter/dark sector can modify N_{eff} :

- Adding a “sterile component.” By “sterile component,” we mean any new degrees of freedom that contribute to radiation and do not inject energy into the Standard Model electromagnetic sector. These can be light dark sector particles or even stochastic gravitational waves. The effect of adding sterile components is summarized in the diagram (3.47) and is constrained by both $\Delta N_{\text{eff}}^{\text{CMB}}$ and $\Delta N_{\text{eff}}^{\text{BBN}}$.

$$\begin{aligned}
\text{Adding sterile components} &\Rightarrow \rho_{\text{rad}} \nearrow \Rightarrow N_{\text{eff}}^{\text{CMB}} \nearrow \\
&\Downarrow \\
H &\nearrow \Rightarrow \text{Less time to burn D} \Rightarrow \text{D/H} \nearrow \\
&\Downarrow \\
&\text{Higher } n/p \text{ freeze-out temperature} \\
&\Downarrow \\
Y_{\text{p}} &\nearrow
\end{aligned} \tag{3.47}$$

To be more concrete, let us consider a dark sector with a real scalar S and a Dirac fermion χ . It interacts with the Standard Model sector purely gravitationally, and the two sectors conserve their entropy separately as they cool. We then have the following relation at a lower temperature,

$$a_{\text{UV}}^3 g_{\text{SM}}^{\text{UV}} T_\gamma^{\text{UV}^3} = a_{\text{IR}}^3 g_{\text{SM}}^{\text{IR}} T_\gamma^{\text{IR}^3}, \quad a_{\text{UV}}^3 g_{\text{D}}^{\text{UV}} T_{\text{D}}^{\text{UV}^3} = a_{\text{IR}}^3 g_{\text{D}}^{\text{IR}} T_{\text{D}}^{\text{IR}^3} \tag{3.48}$$

which can be combined into

$$\left(\frac{T_{\text{D}}}{T_\gamma}\right)_{\text{IR}} = \left(\frac{g_{\text{SM}}^{\text{IR}} g_{\text{D}}^{\text{UV}}}{g_{\text{SM}}^{\text{UV}} g_{\text{D}}^{\text{IR}}}\right)^{1/3} \left(\frac{T_{\text{D}}}{T_\gamma}\right)_{\text{UV}}. \tag{3.49}$$

Therefore, the energy density for the dark sector at the lower temperature is given by

$$\rho_{\text{D}}^{\text{IR}} = g_{\text{D}}^{\text{IR}} \frac{\pi^2}{30} (T_{\text{D}}^{\text{IR}})^4 = g_{\text{D}}^{\text{IR}} \frac{\pi^2}{30} \left(\frac{g_{\text{SM}}^{\text{IR}} g_{\text{D}}^{\text{UV}}}{g_{\text{SM}}^{\text{UV}} g_{\text{D}}^{\text{IR}}}\right)^{4/3} \left(\frac{T_{\text{D}}}{T_\gamma}\right)_{\text{UV}}^4 (T_\gamma^{\text{IR}})^4. \tag{3.50}$$

Consider the UV when all Standard Model particles are in the plasma and the IR before neutrinos decouple. Therefore, we have $g_{\text{SM}}^{\text{UV}} = 106.75$ and $g_{\text{SM}}^{\text{IR}} = 10.75$. For the dark sector, assume $g_{\text{D}}^{\text{UV}} = g_{\text{D}}^{\text{IR}} = g_{\text{D}}$ and that the two sectors have the same temperature at UV, $T_{\text{D}}^{\text{UV}} = T_\gamma^{\text{UV}}$. This yields

$$\rho_{\text{D}}^{\text{IR}} = g_{\text{D}}^{\text{IR}} \frac{\pi^2}{30} \times 0.047 \times (T_\gamma^{\text{IR}})^4 \quad \Rightarrow \quad \Delta N_{\text{eff}} = \frac{\rho_{\text{D}}^{\text{IR}}}{\rho_\nu} = 0.027 g_{\text{D}}. \tag{3.51}$$

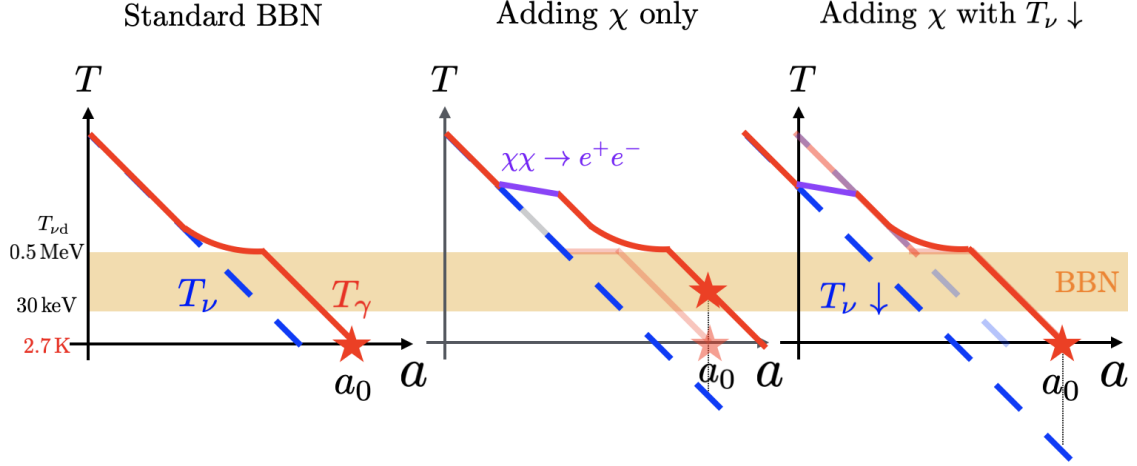


Figure 9. The evolution of the temperature of the neutrino sector T_ν and the Standard Model electromagnetic sector T_γ from before neutrino decoupling to today. Left panel: the evolution in the standard BBN, where T_γ is heated by $(11/4)^{1/3}$ due to electron-positron annihilation. The red star indicates $T_{\gamma,0}$, fixed at 2.7 K. Middle panel: adding χ that contributes to $\chi\chi \rightarrow e^+e^-$. This further heats the Standard Model electromagnetic sector, leading to a higher $T_{\gamma,0}$ than observed. Right panel: To accommodate χ , we need to lower the neutrino temperature so that $T_{\gamma,0} = 2.7$ K.

The number of degrees of freedom of our dark sector (S , χ , and $\bar{\chi}$) is

$$g_D = \underbrace{1}_{S\text{'s dof}} + \underbrace{\frac{7}{8}}_{\text{Fermion discount}} \left(\underbrace{2}_{\chi\text{'s dof}} + \underbrace{2}_{\bar{\chi}\text{'s dof}} \right) = \frac{9}{2} \Rightarrow \Delta N_{\text{eff}} = 0.122. \quad (3.52)$$

Such a dark sector is still allowed by current measurements of $N_{\text{eff}}^{\text{CMB}}$ and $N_{\text{eff}}^{\text{BBN}}$. If we allow asymmetric reheating such that $T_D < T_\gamma$ at UV, ΔN_{eff} from the cold dark sector is further suppressed by $(T_D/T_\gamma)^4$ and it is much easier to escape the N_{eff} constraint.

- A more interesting scenario considers dark sector particles contributing to the electromagnetic sector before BBN. Let us consider a dark matter particle χ with mass $m_\chi > m_e$, which can annihilate into e^+e^- via a dark photon. In this scenario, χ will be in chemical equilibrium with the Standard Model electromagnetic sector at high temperatures. The electromagnetic sector would be heated by $\chi\bar{\chi}$ annihilation when the temperature drops below m_χ . However, this is forbidden as it would lead to a CMB temperature today, $T_{\gamma,0}$, greater than 2.73 K, as shown in the middle panel of Fig. 9.

To accommodate the introduction of χ , we must lower the Standard Model sector temperature before the neutrino decoupling (remember $T_{\gamma,0}$ is the only temperature we measured). This results in a smaller neutrino sector temperature and restores $T_{\gamma,0}$ to the measured value. As summarized in diagram (3.53), there are still two consequences: the lower T_ν leads to lower radiation ρ_{rad} and a lower Hubble parameter during BBN, resulting in a lower $N_{\text{eff}}^{\text{CMB}}$ and lower D/H. The slower expansion of

the universe allows more time for freeze-out and burns more D into heavier elements. Interestingly, Y_p is affected by two opposing factors: the lower n/p freeze-out temperature caused by the smaller Hubble parameter and the smaller n/p interconversion rate induced by the colder neutrinos [49]. The net effect is that Y_p remains almost unchanged compared to the standard BBN.

Add χ ($\chi\chi \rightarrow e^-e^+$) $\Rightarrow$ electromagnetic sector would be heated, but $T_{\gamma,0}$ is fixed

$$\begin{array}{c}
\uparrow \\
T_\nu \searrow \\
\downarrow \\
\rho_{\text{rad}} \searrow \Rightarrow N_{\text{eff}}^{\text{CMB}} \searrow \\
\downarrow \\
H \searrow \Rightarrow \text{More time to burn D} \Rightarrow \text{D/H} \searrow \\
\downarrow \\
\text{Lower } n/p \text{ freeze-out temperature} \\
\text{But } T_\nu \searrow \Rightarrow \text{Less } n/p \text{ interconversion rate} \\
\downarrow \\
Y_p \text{ unchanged}
\end{array} \tag{3.53}$$

One may think of hiding the extra annihilation by adding sterile components. In this case, the changes in $N_{\text{eff}}^{\text{CMB}}$ and D/H cancel. However, after combining diagrams (3.47) and (3.53), the net enhancement in Y_p cannot be mitigated unless additional new physics is introduced. See diagram (3.54). Since Y_p alone is well-measured, we still have strong constraints on χ .

$$\begin{array}{c}
\text{Add } \chi \text{ } (\chi\chi \rightarrow e^-e^+) \text{ and sterile components} \Rightarrow N_{\text{eff}}^{\text{CMB}} \text{ unchanged} \\
\downarrow \\
\text{D/H unchanged, but } Y_p \nearrow
\end{array} \tag{3.54}$$

A more quantitative analysis of the extra annihilation scenario, the combined scenario, or other more involved non-standard BBN scenarios requires implementing the nuclear reaction processes, listed in Fig. 10, using a BBN code. Popular BBN codes include ACROPOLIS [50], ALTERBBN [51], PRIMAT [52, 53], and PRYMORDIAL [54]. Ref. [55] considers a dark sector with a dark photon A' and a complex scalar or Dirac fermion dark matter χ . Using a modified PRIMAT code, the authors obtain $m_\chi > 5.2$ (7.9) MeV for dark matter to be a complex scalar (Dirac fermion), assuming the mass ratio $m_{A'}/m_\chi = 3$. This is shown as the left “walls” in Fig. 11. The bound is valid as long as the coupling between the dark matter and the Standard Model sector is large enough to bring $\chi\chi \leftrightarrow e^+e^-$ to chemical equilibrium when $T_\gamma \gtrsim m_{A'}$, i.e., $(\Gamma_{\chi\chi \leftrightarrow e^+e^-} = H)_{T_\gamma=m_{A'}}$. The result corresponds to $\bar{\sigma}_e \equiv \frac{16\pi\alpha\alpha_D\epsilon^2\mu_{\chi e}^2}{(\alpha^2 m_e^2 + m_{A'}^2)^2} \gtrsim 10^{-43} \text{ cm}^2$.

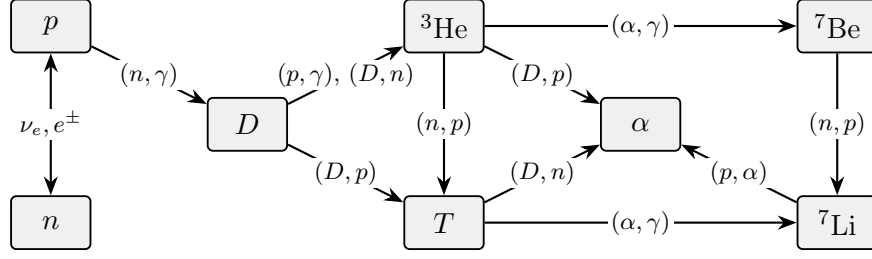


Figure 10. The major nuclear reactions during BBN. An arrow from X to Y labeled (a, b) denotes the reaction $X(a, b)Y$, i.e., $X + a \rightarrow Y + b$; the weak processes interconvert $n \leftrightarrow p$ are $n + e^+ \leftrightarrow p + \bar{\nu}_e$ and $p + e^- \leftrightarrow n + \nu_e$. α : ${}^4\text{He}$, D : ${}^2\text{H}$, T : ${}^3\text{H}$. Cf. [42].

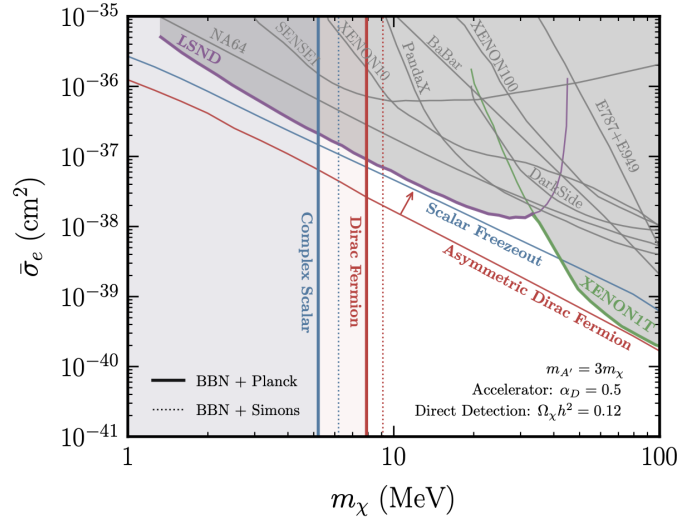


Figure 11. The constraints from N_{eff} of BBN + CMB on dark photon-dark matter model. The plot is taken from [55].

4 Second lecture

There is a blooming of dark matter models in the particle physics community. However, many models lead to similar predictions for astrophysical observations and can be summarized by a few baseline models. Besides CDM, the baseline models are

- Warm dark matter (WDM)
- Fuzzy dark matter (FDM), a.k.a., ultralight dark matter
- Self-interacting dark matter (SIDM)
- Interacting dark matter (IDM)

WDM primarily affects the halo mass function. FDM also alters the halo mass function but, in addition, leads to very different dark matter density profiles within the halos. SIDM does not modify the halo mass function but results in dynamically evolving dark matter

density profiles within the halos. IDM involves strong interactions with baryons, with cross-sections surpassing those of dark matter overburden. IDM can produce a halo mass function similar to WDM and additional features such as dark acoustic oscillations. Some dark matter models combine features from multiple baseline models. An example is atomic dark matter, consisting of dark electrons, protons, and photons. Its phenomenology exhibits characteristics of WDM, SIDM, and IDM.

The popularity of these baseline models is due to two main reasons. Firstly, dark matter on large scales is well-constrained and is expected to be CDM-like. Consequently, significant deviations are anticipated only in the small-scale structures, i.e., dark matter halos. Secondly, astrophysical and cosmological studies often require extensive numerical simulations. Thus, models with a single parameter, such as the dark matter mass for WDM and FDM or the cross-section strength for SIDM, are ideal. New techniques such as machine learning may allow us to thoroughly explore models with multiple parameters, such as IDM and atomic dark matter.

In the following sections, we will discuss two aspects of beyond-CDM imprints on small-scale structures: the effects on structure formation, which influence the distribution of halos of various masses, and the effects on a single halo's density profile.

4.1 The scale-invariant power spectrum from inflation

Where do the primordial density perturbations come from, and why is their spectrum (nearly) scale-invariant? This subsection answers both questions with a minimal amount of quantum field theory in curved spacetime. The presentation follows the lectures of Lam Hui.

4.1.1 Why inflation

Let us restore the curvature term k in the Friedmann equation Eq. (2.12),

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}. \quad (4.1)$$

The observed spatial flatness of the universe requires the curvature term to remain subdominant, so the energy density ρ must decay *slower* than a^{-2} . Using $\rho \propto a^{-3(1+w)}$ (Question 2.3), this requires $3(1+w) < 2$, i.e., $w < -1/3$. The same condition implies accelerated expansion. Multiplying Eq. (4.1) by a^2 and taking a time derivative,

$$\dot{a}^2 = \frac{8\pi G}{3}\rho a^2 - k \quad \Rightarrow \quad 2\dot{a}\ddot{a} = \frac{8\pi G}{3} \frac{d(\rho a^2)}{dt}, \quad (4.2)$$

so ρ decaying slower than a^{-2} gives $d(\rho a^2)/dt > 0$ and hence $\ddot{a} > 0$. *An era with $\dot{a} > 0$ and $\ddot{a} > 0$ is called inflation.*

The canonical case is $w = -1$, for which $\rho = \text{const}$, $H = \text{const}$, and

$$a \propto e^{Ht}, \quad (4.3)$$

i.e., de Sitter space. In practice, this is realized by a scalar field (the inflaton) slowly rolling down a flat potential $V(\phi)$; as long as the potential energy dominates over the kinetic energy, $w \simeq -1$.

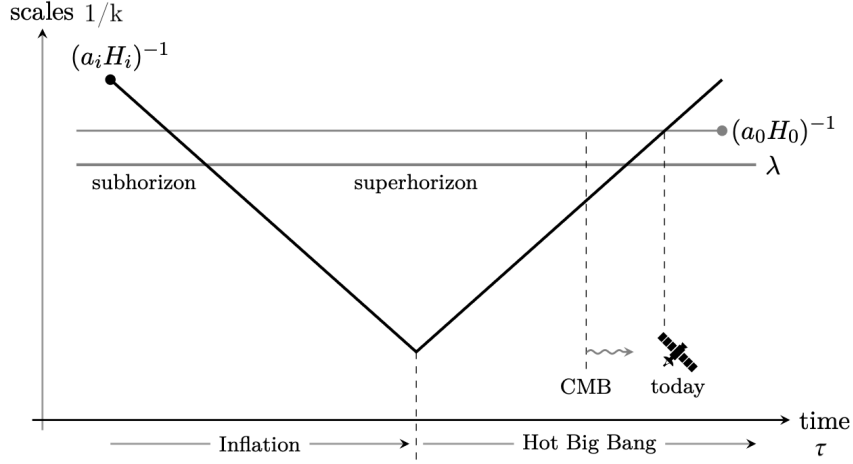


Figure 12. The evolution of comoving scales. The comoving Hubble radius $(aH)^{-1}$ shrinks during inflation and grows during the Hot Big Bang era ($\propto a$ in the radiation era and $\propto a^{1/2}$ in the matter era). A comoving wavelength $\lambda = 1/k$ starts sub-horizon deep in inflation, exits the horizon, stays frozen while super-horizon, and re-enters the horizon during the Hot Big Bang era. The figure is taken from [3].

4.1.2 Goodbye and see you again

The key kinematic feature of inflation is the behavior of the Hubble radius H^{-1} . During inflation, H^{-1} stays constant while physical wavelengths grow, $\lambda_{\text{phy}} \propto a$, so modes *exit* the horizon (“goodbye”). After inflation ends, the Hubble radius grows faster ($H^{-1} \propto a^2$ in the radiation era and $H^{-1} \propto a^{3/2}$ in the matter era, see Sec. 2.2) than λ_{phy} , so modes *re-enter* the horizon (“see you again”). Equivalently, in comoving units, a comoving wavelength λ stays constant while the comoving Hubble radius $(aH)^{-1}$ shrinks during inflation and grows during the Hot Big Bang era, as illustrated in Fig. 12. Whatever amplitude a fluctuation mode acquires when it exits the horizon becomes the initial condition it delivers back when it re-enters. Inflation thus converts sub-horizon quantum fluctuations into super-horizon classical initial conditions.

4.1.3 Warm-up: a massless scalar in de Sitter space

It is convenient to trade the cosmic time t for the conformal time τ , defined by $d\tau = dt/a$,

$$ds^2 = a^2(\tau) (-d\tau^2 + |d\mathbf{x}|^2), \quad a(t) = e^{Ht} \Rightarrow a(\tau) = -\frac{1}{H\tau}. \quad (4.4)$$

The conformal time is negative during inflation: $\tau \rightarrow -\infty$ at early times ($t \rightarrow -\infty$) and $\tau \rightarrow 0^-$ at late times ($t \rightarrow +\infty$). From now on, we use X' to represent a conformal time derivative ($\dot{X}$ represents a normal time derivative). A comoving mode k is characterized by the dimensionless combination

$$|k\tau| = \frac{k}{aH} = \frac{H^{-1}}{\lambda_{\text{phy}}}, \quad (4.5)$$

so $|k\tau| \gg 1$ corresponds to a sub-horizon mode ($\lambda_{\text{phy}} \ll H^{-1}$, early times) and $|k\tau| \ll 1$ to a super-horizon mode ($\lambda_{\text{phy}} \gg H^{-1}$, late times).

As a warm-up, consider a massless *spectator* scalar field ϕ (not the inflaton) in de Sitter space,

$$S = \int d\tau d^3x \sqrt{-g} \left(-\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \right) = \int d\tau d^3x a^4 \left[\frac{1}{2a^2} \left(\frac{\partial \phi}{\partial \tau} \right)^2 - \frac{1}{2a^2} (\nabla \phi)^2 \right]. \quad (4.6)$$

Our goal is the equal-time two-point function $\langle 0 | \phi(\mathbf{x}_1, \tau) \phi(\mathbf{x}_2, \tau) | 0 \rangle$.

Exercise 4.1. Define $\psi \equiv a\phi$. Show, using integration by parts, that the action Eq. (4.6) becomes

$$S = \int d\tau d^3x \frac{1}{2} \left[\psi'^2 - (\nabla \psi)^2 + \frac{2}{\tau^2} \psi^2 \right], \quad (4.7)$$

The field ψ is a canonically normalized scalar in flat spacetime, with a time-dependent mass term inherited from the expansion.

In Fourier space, $\psi(\mathbf{x}, \tau) = \int \frac{d^3k}{(2\pi)^3} \psi_k(\tau) e^{i\mathbf{k} \cdot \mathbf{x}}$, the equation of motion from Eq. (4.7) reads

$$\psi_k'' + \left(k^2 - \frac{2}{\tau^2} \right) \psi_k = 0. \quad (4.8)$$

Two limits are instructive:

- Early times ($|k\tau| \gg 1$): the $2/\tau^2$ term is negligible, and Eq. (4.8) reduces to a Minkowski harmonic oscillator, $\psi_k \propto e^{-ik\tau}$. Deep inside the horizon, we are just doing ordinary flat-space QFT, so we can pick the standard positive-frequency vacuum. This choice is the *Bunch-Davies vacuum*. Physically, we assume inflation had been going on for a while before the largest observable mode exits the horizon, so that all observable modes start deep inside the horizon.
- Late times ($|k\tau| \ll 1$): the k^2 term is negligible, $\psi_k'' \simeq (2/\tau^2) \psi_k = (a''/a) \psi_k$, whose *growing* solution is $\psi_k \propto a$. Hence $\phi_k = \psi_k/a \rightarrow \text{const}$: the field is *frozen* once it is outside the horizon.

The exact solution that interpolating between the two limits is

$$\psi_k(\tau) = e^{-ik\tau} \left(1 - \frac{i}{k\tau} \right). \quad (4.9)$$

Quantizing ψ with these mode functions,

$$\psi(\mathbf{x}, \tau) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2k} \left[e^{-ik\tau + i\mathbf{k} \cdot \mathbf{x}} \left(1 - \frac{i}{k\tau} \right) a_{\mathbf{k}} + e^{+ik\tau - i\mathbf{k} \cdot \mathbf{x}} \left(1 + \frac{i}{k\tau} \right) a_{\mathbf{k}}^\dagger \right], \quad (4.10)$$

with the relativistic normalization $[a_{\mathbf{k}}, a_{\mathbf{k}'}^\dagger] = (2\pi)^3 2k \delta^{(3)}(\mathbf{k} - \mathbf{k}')$ and the Bunch-Davies vacuum defined by $a_{\mathbf{k}}|0\rangle = 0$.

4.1.4 The scale-invariant spectrum

Defining the power spectrum of ϕ through

$$\langle 0 | \phi(\mathbf{x}_1, \tau) \phi(\mathbf{x}_2, \tau) | 0 \rangle = \int \frac{d^3 k}{(2\pi)^3} P_\phi(k, \tau) e^{i\mathbf{k} \cdot (\mathbf{x}_1 - \mathbf{x}_2)}, \quad (4.11)$$

a short computation using the commutator above and $\phi = \psi/a$ gives

$$P_\phi(k, \tau) = \frac{H^2 \tau^2}{2k} \left(1 + \frac{1}{k^2 \tau^2} \right) \xrightarrow{|k\tau| \rightarrow 0} \frac{H^2}{2k^3}. \quad (4.12)$$

At late times, we arrive at

$$\boxed{P_\phi = \frac{H^2}{2k^3} \quad \Rightarrow \quad k^3 P_\phi \text{ is independent of } k.} \quad (4.13)$$

This is what we mean by *scale invariance*. In terms of the dimensionless power spectrum introduced in Sec. 4.2, $\Delta_\phi^2 \equiv k^3 P_\phi / (2\pi^2) = (H/2\pi)^2$: every mode exits the horizon and freezes with the same root-mean-square amplitude, $\phi_{\text{rms}} \sim H/2\pi$, set by the only scale available in de Sitter space, the Hubble rate at horizon exit.

Exercise 4.2. Show the short computation that yields Eq. (4.12).

So far, ϕ has been a spectator. For the inflaton itself, the fluctuations must be treated together with the metric perturbations, starting from

$$S = \int d^4 x \sqrt{-g} \left[\frac{1}{2} M_{\text{pl}}^2 R - \frac{1}{2} g^{\mu\nu} \partial_\mu \phi_{\text{inf}} \partial_\nu \phi_{\text{inf}} - V(\phi_{\text{inf}}) \right], \quad (4.14)$$

writing the metric in the ADM form,

$$ds^2 = -N^2 dt^2 + h_{ij} (dx^i + N^i dt)(dx^j + N^j dt), \quad (4.15)$$

and choosing the unitary gauge, in which the inflaton is unperturbed, $\phi_{\text{inf}}(t, \mathbf{x}) = \phi_{\text{inf}}(t)$, and all fluctuations reside in the spatial metric, $h_{ij} = a^2 e^{2\zeta} \delta_{ij}$. The variable ζ , the co-moving curvature perturbation, is conserved outside the horizon. Carrying out the same quantization for ζ (see [3] for the full derivation) yields the same scale-invariant form with amplitude

$$\Delta_\zeta^2 = \frac{k^3 P_\zeta}{2\pi^2} = \frac{1}{8\pi^2 \epsilon} \left(\frac{H}{M_{\text{pl}}} \right)^2, \quad \epsilon \equiv -\frac{\dot{H}}{H^2}. \quad (4.16)$$

This result can be understood intuitively: the inflaton fluctuation $\delta\phi \sim H/2\pi$ shifts the local end of inflation by $\delta t = \delta\phi/\dot{\phi}_{\text{inf}}$, imprinting a curvature perturbation $\zeta \sim H\delta t = H\delta\phi/\dot{\phi}_{\text{inf}}$.

Because H decreases slowly during inflation ($\epsilon > 0$), modes exiting later (larger k) freeze with slightly smaller amplitudes: the spectrum is not exactly scale-invariant but slightly red-tilted, $\Delta_\zeta^2 \propto k^{n_s-1}$ with $n_s \simeq 0.965$ as measured by Planck [11]. The super-horizon ζ set during inflation provides the adiabatic initial conditions for all species; when a mode re-enters the horizon, ζ seeds the matter overdensity δ_{m} . This is the origin of the scale-invariant initial spectrum, $(k^3 P_{\text{m}})_i \simeq \text{const}$, used in the next subsection.

4.1.5 The spectrum of primordial gravitational waves^{*}

Inflation stretches not only the fluctuations of scalar fields but also those of the metric itself. Consider the transverse-traceless perturbation of the spatial metric (not to be confused with the ADM variable h_{ij} above),

$$ds^2 = -dt^2 + a^2 (\delta_{ij} + h_{ij}) dx^i dx^j, \quad \partial_i h_{ij} = h_{ii} = 0, \quad (4.17)$$

which describes gravitational waves with two polarizations, h_+ and $h_\times$. In the short-wavelength (Isaacson) approximation, the energy density carried by gravitational waves is

$$\rho_{\text{GW}} = \frac{M_{\text{pl}}^2}{4} \sum_{i,j} \langle \dot{h}_{ij}^2(\mathbf{x}) \rangle. \quad (4.18)$$

The amplitude generated during inflation can be estimated using the uncertainty principle. For a mode at horizon crossing, $\dot{h} \sim Hh$, so the gravitational wave energy inside a Hubble patch is

$$\Delta E \sim \rho_{\text{GW}} (H^{-1})^3 \sim \frac{M_{\text{pl}}^2}{4} \langle h^2 \rangle H^2 (H^{-1})^3 = \frac{M_{\text{pl}}^2}{4} \langle h^2 \rangle H^{-1}. \quad (4.19)$$

Treating the mode as a quantum fluctuation that lives for a Hubble time, the uncertainty principle gives $\Delta E \Delta t \sim \Delta E H^{-1} \sim 1$, and therefore

$$\langle h^2 \rangle \sim \frac{4H^2}{M_{\text{pl}}^2}. \quad (4.20)$$

Like the scalar field in the warm-up, the tensor modes freeze at horizon exit with an amplitude set by H .

The formal computation requires no new machinery. Expanding the Einstein-Hilbert action to second order in h_{ij} shows that each polarization is a massless scalar field in disguise, with the canonical normalization $\phi^s = (M_{\text{pl}}/2) h^s$ for $s = +, \times$. Each polarization therefore inherits the scale-invariant spectrum of the warm-up, $\Delta_\phi^2 = (H/2\pi)^2$, and summing over the two polarizations yields

$$\Delta_h^2(k) = 2 \times \left(\frac{2}{M_{\text{pl}}} \right)^2 \left(\frac{H}{2\pi} \right)^2 = \frac{2}{\pi^2} \left(\frac{H}{M_{\text{pl}}} \right)^2, \quad (4.21)$$

consistent with the uncertainty-principle estimate up to an $\mathcal{O}(1)$ factor. See [3] for the full derivation.

Unlike the scalar amplitude, which depends on the combination H^2/ϵ , the tensor amplitude depends on H alone. A detection of Δ_h^2 would therefore directly measure the Hubble rate during inflation, and hence the energy scale of inflation, $V^{1/4} \simeq (3H^2 M_{\text{pl}}^2)^{1/4}$. The tensor amplitude is conventionally quoted as the tensor-to-scalar ratio,

$$r \equiv \frac{\Delta_h^2}{\Delta_\zeta^2} = 16\epsilon. \quad (4.22)$$

The current bound from BICEP/Keck and Planck, $r < 0.036$ at 95% CL [56], corresponds to $V^{1/4} \lesssim 1.4 \times 10^{16}$ GeV, tantalizingly close to the scale of grand unification.

4.2 Small-scale power spectrum: probing dark matter in the early universe

Our universe today is full of matter-rich halos and matter-poor voids, which respectively developed from small initial overdensities and underdensities in the primordial matter distribution. These overdensities, denoted by

$$\delta_{\text{m}}(\mathbf{x}) \equiv \frac{\rho_{\text{m}}(\mathbf{x}) - \bar{\rho}_{\text{m}}}{\bar{\rho}_{\text{m}}}, \quad (4.23)$$

where $\bar{\rho}_{\text{m}}$ is the averaged matter density, grow over time due to gravity. Specifically, they first grew in the linear perturbation regime, where $\delta_{\text{m}} \ll 1$, as the long-wavelength modes re-entered the horizon. As these overdensities evolved, they entered the non-linear regime ($\delta_{\text{m}} > 1$), leading to the collapse of surrounding background dark matter and the formation of virialized halos. The average density of a dark matter halo is approximately $\Delta = 200$ times the background density. Smaller halos merged or accreted matter to form larger ones, a process manifested in the halo mass-redshift relation and the concentration-halo mass-redshift relation, with concentration indicating how compact a halo is.

In practice, we use “two-point statistics,” i.e., the matter power spectrum and the halo mass function, to characterize the distributions of overdensities and halos, respectively. These tools are essential for studying the properties of dark matter and beyond. While large-scale observations generally support the predictions of Λ CDM, there can be significant deviations on smaller scales. Additionally, observations of galaxies and dwarf galaxies have raised several challenging questions for Λ CDM, such as the diversity of rotation curves, the too-big-to-fail problem, and many ultra-dense substructures recently discovered in lensing and stellar kinematics. They all hint at the need for dark matter models beyond CDM.

As with other types of fields, it is more convenient to analyze the overdensity field in momentum space than in position space. The two representations are related by the Fourier transform:

$$\delta_{\text{m}}(\mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \delta_{\text{m}}(\mathbf{k}) e^{-i\mathbf{k}\cdot\mathbf{x}}, \quad (4.24)$$

where $\mathbf{k}$ is the 3D momentum vector. The matter power spectrum, P_{m} , is related to the two-point correlation function of $\delta_{\text{m}}(\mathbf{k})$ by:

$$\langle \delta_{\text{m}}(\mathbf{k}) \delta_{\text{m}}(\mathbf{k}') \rangle = (2\pi)^3 \delta^3(\mathbf{k} + \mathbf{k}') P_{\text{m}}(k) \quad (4.25)$$

where $\langle \dots \rangle$ averages over all the spatial configurations and P_{m} is only a function of the magnitude of $\mathbf{k}$ due to the cosmological principle. Often it is convenient to introduce the dimensionless matter power spectrum $\Delta^2(k)$ that relates to $P_{\text{m}}(k)$ by

$$\Delta^2(k) \equiv \frac{k^3 P_{\text{m}}(k)}{2\pi^2}. \quad (4.26)$$

With the Fourier convention above, $\Delta^2(k)$ measures the contribution to the density variance per logarithmic interval in k , i.e., $\langle \delta_{\text{m}}^2(\mathbf{x}) \rangle = \int d \ln k \Delta^2(k)$.

Fig. 13 (Fig. 14) shows the (dimensionless) linear matter power spectrum at $z = 0$ for the CDM, WDM, FDM, and IDM, along with observations from CMB and LSS. As

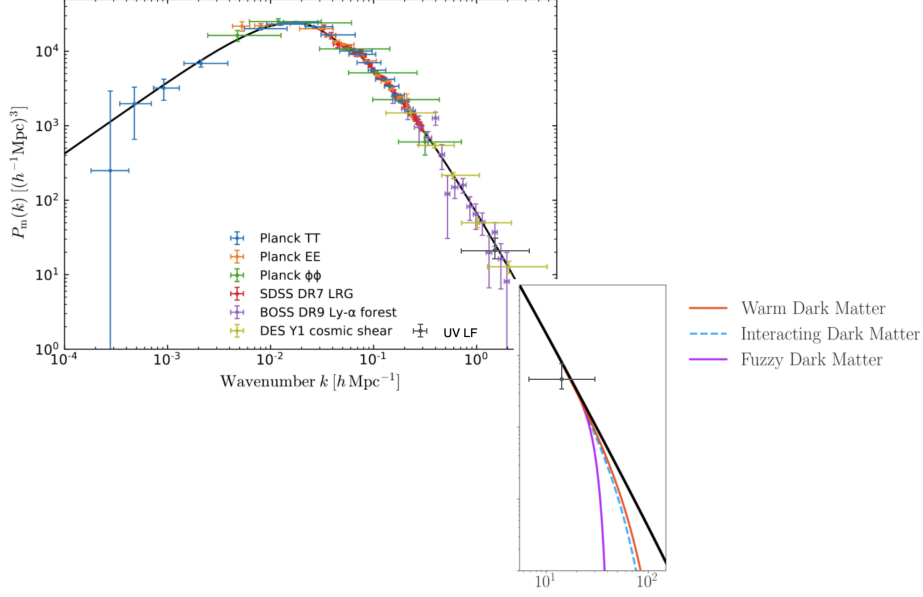


Figure 13. Linear matter power spectrum at $z = 0$, reproduced from Nadler’s BCCP seminar slides [57]. We added constraints from the ultraviolet galaxy luminosity function (UV LF) [58]. The measured data are compiled by [59]. From the largest scales to smaller ones, the data includes Planck CMB, galaxy clustering, galaxy weak lensing, Ly- α clustering, and UV LF.

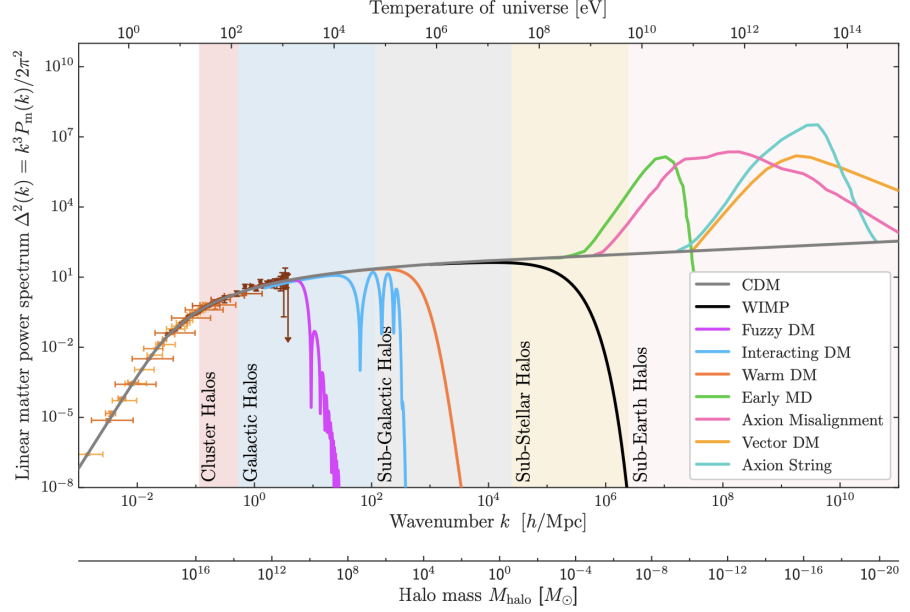


Figure 14. Dimensionless linear matter power spectrum at $z = 0$, reproduced from [60]. A detailed explanation of model predictions can be found in [60]. The measured data are compiled by [59].

is common in astronomical plots, the data should be read from right to left. Modes with smaller wavenumbers enter the horizon earlier than those with larger wavenumbers. First, let us focus on the CDM prediction, which sets the baseline of the matter power spectrum

that agrees well with observations at $k \lesssim 10 h \text{ Mpc}^{-1}$.

4.2.1 The power spectrum of CDM

A notable feature of the CDM power spectrum is the turnover around $k = 0.02 h \text{ Mpc}^{-1}$. This scale corresponds to modes entering the horizon at the matter-radiation equality, denoted as k_{eq} . For modes with wavenumber $k < k_{\text{eq}}$, the power spectrum behaves as $P_{\text{m}}(k) \propto k$. For modes with wavenumber $k > k_{\text{eq}}$, the relationship changes to $P_{\text{m}}(k) \propto k^{-3}$. In the dimensionless spectrum, where $P_{\text{m}}(k)$ is multiplied by k^3 , we find $\Delta^2(k) \propto k^4$ for $k < k_{\text{eq}}$, and $\Delta^2(k)$ exhibits a logarithmic-like growth with respect to k for $k > k_{\text{eq}}$.

The turnover behavior can be well explained using the linear perturbation theory of the Einstein-Boltzmann equation. A more detailed and comprehensive derivation can be found in textbooks such as [3], which starts by perturbing a set of fluid dynamics equations:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0, \quad \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla \Phi - \frac{\nabla P}{\rho}, \quad \nabla^2 \Phi = 4\pi G \rho, \quad (4.27)$$

where ρ , P , and $\mathbf{u}$ are the fluid's density, pressure, and velocity, respectively, and Φ is the gravitational potential. Here, we simplify the derivation by directly introducing the resulting master equation:

$$\frac{d^2 \delta_{\text{m}}}{dt^2} + 2H \frac{d\delta_{\text{m}}}{dt} = 4\pi G \bar{\rho}(t) \delta_{\text{m}}, \quad (4.28)$$

where we assume that matter has zero pressure and all matter clumps. Although this assumption is not entirely accurate (e.g., cosmic neutrinos do not clump), it is not far off and simplifies our derivation. To solve the master equation, we use the ansatz

$$\delta_{\text{m}}(\mathbf{k}, t) = \delta_{+}(\mathbf{k}) D_{+}(t) + \delta_{-}(\mathbf{k}) D_{-}(t), \quad (4.29)$$

to separate the $\mathbf{k}$ -dependent and t -dependent parts of δ_{m} . The subscript “+” refers to the growing mode, and “−” to the decaying mode. We will focus on the growing mode, which leads to the large matter perturbation in the later universe.

In the matter era, we have $a \propto t^{2/3}$, $H = 2/(3t) \propto a^{-3/2}$, and $4\pi G \bar{\rho}_{\text{m}}(t) = 3H^2/2 = 2/(3t^2)$ (see Tab. 1 for a summary). Substituting those relations into Eq. (4.28) yields

$$\frac{d^2 D}{dt^2} + \frac{4}{3t} \frac{dD}{dt} = \frac{2}{3t^2} D \quad \Rightarrow \quad D(t) = c_1 t^{-1} + c_2 t^{2/3}, \quad (4.30)$$

where c_i 's are constants. From the solution, we find that the growing mode gives $D_{+} \propto t^{2/3} \propto a$.

In the radiation era, the scale factor behaves as $a \propto t^{1/2}$, and the Hubble parameter is given by $H = 1/(2t) \propto a^{-2}$. In such a universe,

$$\ddot{\delta}_{\text{m}} \sim H \dot{\delta}_{\text{m}} \sim H^2 \delta_{\text{m}} \sim 4\pi G (\bar{\rho}_{\text{rad}} + \bar{\rho}_{\text{m}}) \delta_{\text{m}} \gg 4\pi G \bar{\rho}_{\text{m}} \delta_{\text{m}}, \quad (4.31)$$

so the right-hand side of Eq. (4.28) can be approximated as zero:

$$\frac{d^2 D}{dt^2} + \frac{1}{t} \frac{dD}{dt} = 0 \quad \Rightarrow \quad D(t) = c_1 + c_2 \ln t. \quad (4.32)$$

Therefore, the growing mode in this era gives $D_+ \propto \ln t \propto \ln a$.

The above results suggest that sub-horizon fluctuations in the radiation era only grow logarithmically, or approximately $\delta_m \sim \text{const}$, while in the matter era, the growth accelerates to $\delta_m \propto a$. This difference in growth rates leads to the k -dependence in the matter power spectrum. This is because different k -modes re-enter the horizon at different times in different eras, experiencing different sub-horizon growth. Let us look at it in more detail.

Inflation models suggest that the initial matter power spectrum is scale-invariant. We will discuss this in Sec. 4.1. This fact means

$$(k^3 P_m)_i = \text{const.} \quad (4.33)$$

Modes outside the horizon always have the same shape as the initial spectrum. Now consider modes re-entering the horizon during the matter era: their amplitudes grow by (a_0/a_h) , where a_0 is the scale factor today (ignoring the dark energy era) and a_h is the scale factor at horizon re-entering. In the matter era, a_h is determined by

$$k = a_h H(a_h) \propto a_h a_h^{-3/2} \Rightarrow a_h \propto k^{-2}. \quad (4.34)$$

Therefore, the power spectrum is scaled as

$$k^3 P_m \propto \left(\frac{a_0}{a_h}\right)^2 (k^3 P_m)_i \propto k^4. \quad (4.35)$$

The square over (a_0/a_h) is because $P_m \propto \delta_m^2$.

For modes crossing the horizon during the radiation era, their amplitudes grow by $\ln(a_{\text{eq}}/a_h)$, where a_{eq} is the scale factor at matter-radiation equality. a_h in the radiation era is given by

$$k = a_h H(a_h) \propto a_h a_h^{-2} \Rightarrow a_h \propto k^{-1}. \quad (4.36)$$

This leads to the power spectrum scaling

$$k^3 P_m \propto \left(\ln \frac{a_{\text{eq}}}{a_h}\right)^2 (k^3 P_m)_i \propto \left(\ln \frac{k}{k_{\text{eq}}}\right)^2. \quad (4.37)$$

Note that when those modes later evolve in the matter era, their amplitudes will be enhanced by a common factor that does not affect the shape of the spectrum. Combining the above results, we have the k -dependence

$$P_m(k) \propto \begin{cases} k & k < k_{\text{eq}}, \\ k^{-3} \left(\ln \frac{k}{k_{\text{eq}}}\right)^2 & k > k_{\text{eq}}. \end{cases} \quad (4.38)$$

The evolution of the power spectrum's shape for different cosmological eras is depicted in Fig. 15.

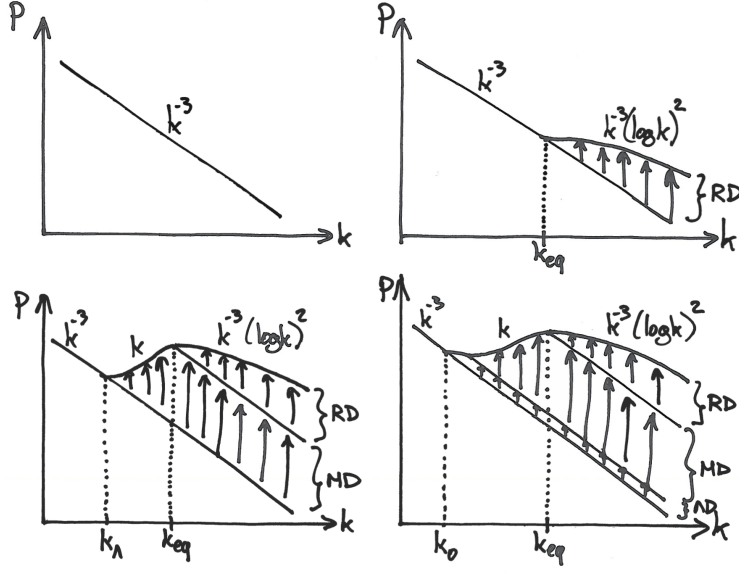


Figure 15. How the shape of the matter power spectrum evolves in time. The four panels correspond to the initial spectrum, the spectra at the radiation era, matter era, and dark energy era, respectively. The figure is taken from [61].

4.2.2 The power spectrum of dark matter beyond CDM

Many dark matter models beyond CDM can suppress the power spectrum at small scales. Because the matter power spectrum $P_m(k)$ is related to the halo mass function $\frac{dN}{dM_h}(M_h)$, by carefully observing galaxies and dwarf galaxies, we have an opportunity to detect deviations from the CDM model and learn more about the properties of dark matter. In addition, because the small-scale structures began to grow in the radiation era, new physics that suppressed them should be active during that period. Therefore, the small-scale structure provides us with a window to probe dark matter in the early universe.

Now, let us focus on a few examples:

1. Warm Dark Matter: The comoving wavelength for free-streaming is given by

$$\lambda_{\text{fs}} = \int_{t_i}^{t_{\text{eq}}} dt \frac{\langle v(t) \rangle}{a(t)}, \quad k_{\text{fs}} = \frac{2\pi}{\lambda_{\text{fs}}}, \quad (4.39)$$

where $\langle v(t) \rangle$ is the physical velocity averaged over the WDM's phase space. Note that the integral extends from an initial time t_i to the matter-radiation equality. Beyond the matter-radiation equality, the Jeans length drops dramatically, and perturbations could collapse under gravity. A crude estimation finds [62]

$$\lambda_{\text{fs}} = \int_0^{t_{\text{NR}}} dt \frac{c}{a(t)} + \int_{t_{\text{NR}}}^{t_{\text{eq}}} dt \frac{c a_{\text{NR}}}{a^2(t)} = r_H(t_{\text{NR}}) \left(1 + \frac{1}{2} \ln \frac{t_{\text{eq}}}{t_{\text{NR}}} \right), \quad (4.40)$$

where $r_H(t_{\text{NR}})$ is the comoving horizon at t_{NR} . The WDM moves at the speed of light c before t_{NR} and slows down as a^{-1} between t_{NR} and t_{eq} . Choosing t_{NR} to be the

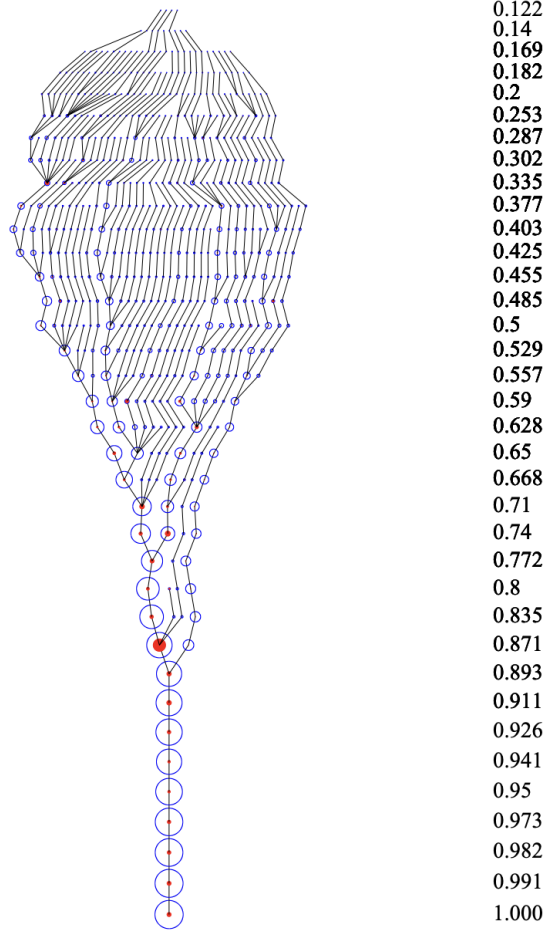


Figure 16. Hierarchical structure formation [63]. The diagram illustrates the merging history of a cluster-size dark matter halo at $a = 1$ from smaller progenitor halos. The numbers indicate the scale factor a .

time when the temperature drops to $T_{\text{WDM}} = m_{\text{WDM}}/3$ yields

$$\lambda_{\text{fs}} = 0.4 \left(\frac{m_{\text{WDM}}}{\text{keV}} \right)^{-4/3} \left(\frac{\Omega_{\text{WDM}} h^2}{0.135} \right)^{1/3} h^{-1} \text{Mpc}. \quad (4.41)$$

This free-streaming comoving wavelength sets a cutoff on the dark matter halo mass:

$$M_{\text{cut}} = \frac{4\pi}{3} \rho \left(\frac{\pi}{k_{\text{fs}}} \right)^3 = \frac{4\pi}{3} \rho \left(\frac{\lambda_{\text{fs}}}{2} \right)^3, \quad (4.42)$$

where ρ is the dark matter density. Below this cutoff, we do not expect to observe any halos because free-streaming washes out the density perturbations.

Stronger constraints on WDM can be obtained by realizing that the removal of smaller halos can affect the properties of larger halos with masses above M_{cut} , causing a departure from the observed halo mass function. This is because larger halos grow out of smaller halos according to hierarchical structure formation (see Fig. 16 and Sec. 5.2).

The halo mass function is closely related to the linear matter power spectrum. However, the exact relation is non-trivial, and one needs to run cosmological simulations to pinpoint it. Fortunately, WDM is one of the beyond-CDM models that has been intensively studied in the simulation community (thanks to the resolved “missing satellite problem”). Based on simulation results, Refs. [64, 65] find that the power spectrum of WDM relative to CDM is given by

$$\frac{P_{\text{WDM}}(k)}{P_{\text{CDM}}(k)} = [1 + (\alpha k)^{2\nu}]^{-\frac{10}{\nu}}, \quad (4.43)$$

with $\nu = 1.12$ and

$$\alpha = 0.049 \left(\frac{m_{\text{WDM}}}{\text{keV}} \right)^{-1.11} \left(\frac{\Omega_{\text{WDM}}}{0.25} \right)^{0.11} \left(\frac{h}{0.7} \right)^{1.22} h^{-1} \text{Mpc}, \quad (4.44)$$

where m_{WDM} and Ω_{WDM} are the WDM mass and abundance, respectively. Also note that

$$\frac{P_{\text{beyond CDM}}(k)}{P_{\text{CDM}}(k)} = [1 + (\alpha k)^\beta]^\gamma, \quad (4.45)$$

is the common functional form used to parameterize the power spectrum for beyond-CDM dark matter, which guarantees $P_{\text{beyond CDM}}/P_{\text{CDM}} = 1$ for $k \rightarrow 0$.

2. Fuzzy dark matter: FDM is naturally associated with the de Broglie wavelength scale of the ultralight dark matter,

$$\lambda_{\text{dB}} = \frac{1}{m_{\text{FDM}} v}. \quad (4.46)$$

This scale is linked to the soliton size at the center of an FDM halo. We are interested in the gravitational instability scale and the halo mass cutoff of FDM, which are naturally associated with the Jeans scale of FDM. The derivation of the Jeans scale starts from the fluid description of FDM, also known as the Madelung equations [66],

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0, \quad \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla \Phi + \frac{\hbar^2}{2m_{\text{FDM}}^2} \nabla \left(\frac{\Delta \sqrt{\rho}}{\sqrt{\rho}} \right), \quad \nabla^2 \Phi = 4\pi G(\rho - \bar{\rho}), \quad (4.47)$$

where we assume FDM is pressureless. Perturbing these equations around a homogeneous distribution ($\rho = \bar{\rho}$, $\mathbf{v} = 0$, $\Phi = 0$) yields,

$$\frac{\partial^2}{\partial t^2} \delta_{\text{m}} = 4\pi G \rho \delta_{\text{m}} - \frac{\hbar^2 k^4}{4m_{\text{FDM}}^2} \delta_{\text{m}}. \quad (4.48)$$

This equation is similar to Eq. (4.28) except for the absence of Hubble friction and the appearance of the “quantum pressure” term. The solution of Eq. (4.48) is given by

$$\delta_{\text{m}}(t) = c_1 \exp \left(t \sqrt{4\pi G \rho - \frac{\hbar^2 k^4}{4m_{\text{FDM}}^2}} \right) + c_2 \exp \left(-t \sqrt{4\pi G \rho - \frac{\hbar^2 k^4}{4m_{\text{FDM}}^2}} \right) \quad (4.49)$$

The Jeans wavenumber corresponds to the critical value of k that allows the perturbation δ_m to have a growing mode. It is set by

$$k_J = \frac{2\pi^{1/4} G^{1/4} \rho^{1/4} m_{\text{FDM}}^{1/2}}{\hbar^{1/2}}, \quad \lambda_J = \frac{2\pi}{k_J}. \quad (4.50)$$

Modes with $k > k_J$ are suppressed due to the quantum pressure. The halo cutoff mass is given by [7]

$$M_{\text{cut}} = \frac{4}{3} \pi \rho \left(\frac{\lambda_J}{2} \right)^3 = 1.5 \times 10^7 M_\odot \left(\frac{\Omega_{\text{FDM}}}{0.27} \right)^{1/4} \left(\frac{h}{0.7} \right)^{1/2} \left(\frac{10^{-22} \text{ eV}}{m_{\text{FDM}}} \right)^{3/2} \quad (4.51)$$

Similar to the WDM case, stronger constraints can be obtained by considering the departure from the CDM power spectrum, i.e., the imprint of FDM on larger halos. Using an emulator, analysis of the Lyman- α forest constrains $m_{\text{FDM}} > 2 \times 10^{-20}$ eV [67].

3. Interacting Dark Matter: The gravitational instability scale for IDM is determined by the horizon size, R_H , at the moment when dark matter and Standard Model particles kinetically decouple.

$$\lambda_{\text{kd}} = R_H(\mathcal{R}_{\text{kd}} = H), \quad (4.52)$$

where $\mathcal{R}_{\text{kd}}$ is the kinetic decoupling rate, and we use $\chi - p$ interactions as an example to illustrate the derivation. Similar kinetic decoupling processes apply to other dark matter-Standard Model interactions, such as $\chi - e$ interactions. Only modes with wavelengths greater than λ_{kd} can gravitationally collapse to form halos. Modes with $\lambda < \lambda_{\text{kd}}$ are suppressed due to the strong interactions which transfer extra random motions from the Standard Model fluid to the dark matter fluid. For a single $2 \rightarrow 2$ elastic scattering process, the dark matter particle with velocity $\mathbf{v}_\chi$ will experience a velocity change of

$$\Delta \mathbf{v}_\chi = \frac{m_p}{m_\chi + m_p} v_{\text{rel}} \left(\hat{\mathbf{n}}_\chi - \frac{\mathbf{v}_{\text{rel}}}{v_{\text{rel}}} \right), \quad (4.53)$$

where $\hat{\mathbf{n}}_\chi$ is the direction of the scattered dark matter particle in the center-of-mass frame. The peculiar velocity of the dark matter fluid, $\mathbf{V}_\chi$, is then given by

$$\frac{d}{dt} \mathbf{V}_\chi = n_\chi \langle \sigma_{\chi p \rightarrow \chi p} v_{\text{rel}} \Delta \mathbf{v}_\chi \rangle = -\mathcal{R}_{\text{kd}} (\mathbf{V}_\chi - \mathbf{V}_p), \quad \mathcal{R}_{\text{kd}} \simeq \frac{m_p}{m_\chi + m_p} n_p \sigma_{\chi p \rightarrow \chi p} v_{\text{rel}} \quad (4.54)$$

where $\langle \cdot \rangle$ denotes averaging over all scattering angles and the velocity distributions of dark matter and protons.

The complete derivation of λ_{kd} is more involved. Dark matter and protons are initially fully coupled, then thermally decoupled, and finally kinetically decoupled. During the period between thermal and kinetic decoupling, the temperature of dark matter evolves as $T_\chi \propto a^{-2}$, which differs from the Standard Model plasma's evolution $T_\gamma \propto a^{-1}$. Therefore, it is necessary to first determine the scale factor at thermal decoupling

a_{td} and then link it to the scale factor at kinetic decoupling a_{kd} . For more details, see [68].

Again, better constraints can be obtained through cosmological simulations starting from the IDM's initial linear power spectrum. However, this involves running simulations to scan over m_χ and $\sigma_{\chi p}$ and can be costly. An alternative way to constrain IDM is by comparing its power spectrum to that of WDM with the maximally allowed m_{WDM} . Such a procedure is allowed because the spectra of WDM and IDM are similar. One can constrain IDM by requiring $P_{\text{IDM}}(m_{\text{IDM}}, \sigma_{\chi p})/P_{\text{CDM}}$ to always lie below $P_{\text{WDM}}(m_{\text{WDM}}^{\text{limit}})/P_{\text{CDM}}$. For more details, see [68].

4.2.3 Observations

There is a correspondence between modes with comoving wavenumber k and the mass of the halo systems, which follows the relation:

$$M_{\text{h}} \simeq \frac{4\pi}{3} \rho_{\text{m}} \left(\frac{\pi}{k} \right)^3 \simeq 10^{10} M_{\odot} \left(\frac{10 h \text{ Mpc}^{-1}}{k} \right)^3. \quad (4.55)$$

Thus, we have three important observational targets for different values of k :

1. $k \sim 0.1 - 1 h \text{ Mpc}^{-1}$: galaxy cluster halos;
2. $k \sim 1 - 100 h \text{ Mpc}^{-1}$: galaxy halos;
3. $k \gtrsim 100 h \text{ Mpc}^{-1}$: completely dark halos whose gravitational potential is too shallow to form galaxies.

Probing the matter power spectrum at small scales requires finding methods to robustly investigate dark matter halos at those scales. For galaxy cluster halos, we can probe them by tracing the hydrogen distribution using the Lyman- α forest, where light from distant quasars is absorbed by intergalactic neutral hydrogen residing inside halos. For galaxy halos, we can infer their abundance by observing luminous Milky Way satellite galaxies through the “galaxy-halo connection”. For completely dark halos, we must rely on other methods, such as strong gravitational lensing or stellar streams.

To showcase the potential of different probes, we present the limits on WDM inferred from various methods or their combinations in Tab. 3 and show the parameter space constraints on IDM in Fig. 17 (taken from [68]). As shown in Fig. 17, the resulting constraints from the Milky Way satellite and Lyman- α on $\sigma_{\chi p(\chi e)}$ cover a wide range of m_χ . It rules out almost all large values for the dark matter overburden scenario, where the null detection of dark matter in underground direct detection experiments is due to the strong interactions between dark matter and the Earth's atmosphere and rocks. The result clearly shows that the cosmological probes complement direct detection experiments and other dark matter-dark sector probes.

Ref.	Data type	$m_{\text{WDM}}^{95\% \text{C.L.}} [\text{keV}]$	h	Ω_{dm}
[69]	luminous satellite galaxies	6.5	0.7	0.24
[70]	stellar stream, luminous satellite galaxies	6.3	0.67	0.26
[71]	strong lensing, luminous satellite galaxies, Lyman- α	6.733	0.7	0.25

Table 3. 95% C.L. lower limits on the WDM mass (3rd column) from MW subhalo data reported in the literature. The 2nd, 4th, and 5th columns respectively show the data type, the value of h , and the value of Ω_{dm} that was used in the derivation of the limits.

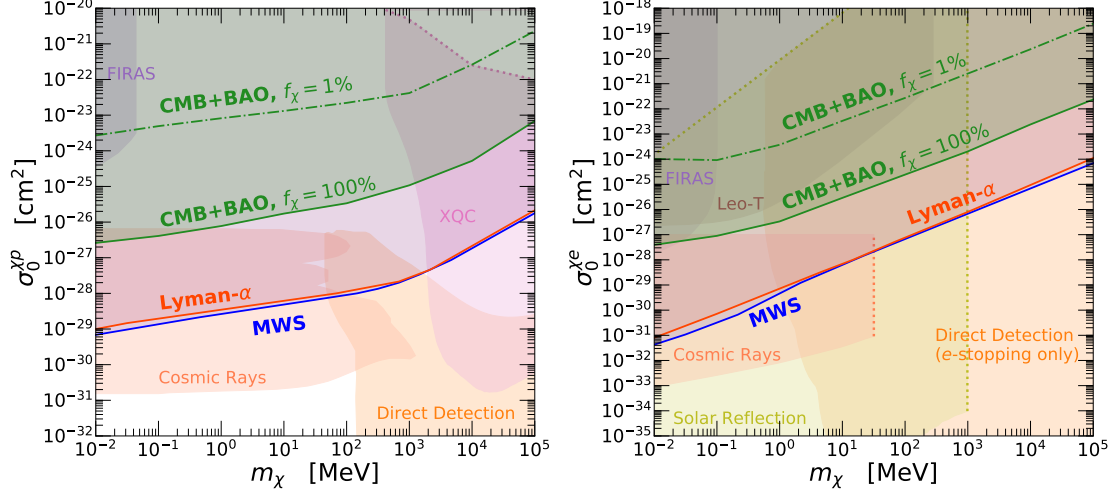


Figure 17. 95% C.L. bounds from MW subhalos (solid blue), Lyman- α forest (solid red), CMB+BAO (solid green) datasets, for velocity-independent (left) dark matter–protons and (right) dark matter–electron interactions. We also include constraints from previous literature. For dark matter–proton interactions (left): XQC rocket (pink) [72–74], various direct-detection experiments (orange: CRESST-III [75, 76], CRESST surface run [77], XENON1T [78], and Migdal effect-based EDELWEISS [79]), and cosmic-ray accelerated dark matter (coral red: Mini-BOONE and XENON1T [81]; and Daya Bay and KamLAND [82]). For dark matter–electron interactions (right): gas cooling of Leo-T dwarf galaxy (brown) [83], various direct-detection experiments as considered in [84] (orange: XENON10 [85–87], XENON100 [86, 88], DarkSide-50 [89], CDMS-HVeV [90], protoSENSEI [91, 92], and SENSEI at MINOS [93]); cosmic-ray accelerated dark matter (coral red: Super-K and MiniBooNE [82, 94]), and bounds from solar reflection (yellow) [95, 96]. The dotted lines represent our conservative assumptions about the limits of the XQC (pink, with the ceiling from [73]); cosmic-ray (coral red [94]), and solar reflection (yellow [96]) bounds. In both panels: FIRAS (purple) [97–99].

5 Third lecture

Previously, we are working in the linear growth regime, where gravity is weak. Here we will focus on the nonlinear regime, where gravity is strong, under which, dark matter forms halos and halos, through hierarchical growth, forms larger halos.

5.1 The formation of dark matter halos

How does a small overdensity grow into a virialized dark matter halo? The linear theory of Sec. 4.2 applies only while $\delta_m \ll 1$. To follow a perturbation into the non-linear regime, we use the classic idealization of the *spherical collapse* model [100]. We first follow a single mass shell, which yields two famous numbers, $\delta_c \simeq 1.686$ and $\Delta_{\text{vir}} = 18\pi^2 \simeq 178$. We then follow the continuous accretion of many shells (the *secondary infall*), which yields self-similar halo profiles and a physically motivated halo boundary, the *splashback radius*.⁵

5.1.1 Spherical collapse of a single mass shell

Consider a spherical overdense region in an Einstein-de Sitter (EdS) universe, i.e., a flat, matter-dominated universe. In reality, EdS universe occurs when the radiation is sufficiently diluted away, i.e., $z < 100$. In such a universe, a mass shell at radius r enclosing mass M obeys

$$\frac{d^2 r}{dt^2} = -\frac{GM(r, t)}{r^2}. \quad (5.1)$$

Before any shells cross, M is constant in time, and Eq. (5.1) integrates to the energy conservation equation

$$\frac{1}{2} \left(\frac{dr}{dt} \right)^2 - \frac{GM}{r} = -\frac{GM}{r_{\text{max}}}, \quad (5.2)$$

where the constant is fixed by the turnaround radius r_{max} , at which $dr/dt = 0$. The bound trajectory is a cycloid, conveniently written in terms of the *development angle* θ :

$$r = \frac{r_{\text{max}}}{2} (1 - \cos \theta), \quad t = \frac{t_{\text{max}}}{\pi} (\theta - \sin \theta), \quad t_{\text{max}} = \frac{\pi}{2} \sqrt{\frac{r_{\text{max}}^3}{2GM}}. \quad (5.3)$$

The shell first expands with the Hubble flow ($\theta \ll 1$), turns around at $\theta = \pi$ ($r = r_{\text{max}}$, $t = t_{\text{max}}$), and formally collapses to $r = 0$ at $\theta = 2\pi$, $t_{\text{col}} = 2t_{\text{max}}$.

Matching the cycloid to the initial conditions relates $(r_{\text{max}}, t_{\text{max}})$ to the initial perturbation. We specify the initial data at a time t_i deep in the matter era. The shell starts at a radius r_i and encloses the mass

$$M = \frac{4\pi}{3} \rho_b(t_i) (1 + \delta_i) r_i^3 = \frac{2r_i^3}{9Gt_i^2} (1 + \delta_i), \quad (5.4)$$

where $\delta_i \ll 1$ is the initial mean overdensity within the shell, and we used $\rho_b(t_i) = 1/(6\pi Gt_i^2)$ (see Tab. 1). For the initial velocity, we take, for simplicity, the unperturbed Hubble flow,

$$\left. \frac{dr}{dt} \right|_{t_i} = H(t_i) r_i = \frac{2r_i}{3t_i}. \quad (5.5)$$

Evaluating the specific energy of the shell, the left-hand side of Eq. (5.2), with the initial data gives

$$E = \frac{1}{2} \left(\frac{2r_i}{3t_i} \right)^2 - \frac{GM}{r_i} = \frac{2r_i^2}{9t_i^2} - \frac{2r_i^2}{9t_i^2} (1 + \delta_i) = -\frac{2r_i^2}{9t_i^2} \delta_i. \quad (5.6)$$

⁵This subsection is based on the author's notes on the splashback radius and has been cross-checked against Wayne Hu's Astro 321 lecture notes (Set 7: Spherical Collapse & Halo Model).

The kinetic and potential contributions cancel up to the perturbation: it is the small initial overdensity that renders the shell bound. Equating Eq. (5.6) with $E = -GM/r_{\max}$, and then using the expression for $t_{\max}$ in Eq. (5.3), we obtain, to leading order in δ_i ,

$$r_{\max} = \frac{r_i}{\delta_i}, \quad t_{\max} = \frac{3\pi}{4} t_i \delta_i^{-3/2}. \quad (5.7)$$

The smaller the initial overdensity, the later and the farther out the shell turns around.

Next, let us extract the early-time behavior of the overdensity. Expanding Eq. (5.3) for $\theta \ll 1$,

$$r = \frac{r_{\max}}{4} \theta^2 \left(1 - \frac{\theta^2}{12}\right) + \mathcal{O}(\theta^6), \quad t = \frac{t_{\max}}{6\pi} \theta^3 \left(1 - \frac{\theta^2}{20}\right) + \mathcal{O}(\theta^7), \quad (5.8)$$

inverting the second relation, $\theta = (6\pi t/t_{\max})^{1/3} \left[1 + \frac{1}{60} (6\pi t/t_{\max})^{2/3} + \dots\right]$, and substituting it into the first, we find

$$r(t) = \frac{r_{\max}}{4} \left(\frac{6\pi t}{t_{\max}}\right)^{2/3} \left[1 - \frac{1}{20} \left(\frac{6\pi t}{t_{\max}}\right)^{2/3} + \dots\right]. \quad (5.9)$$

At leading order, $r \propto t^{2/3} \propto a$: the shell simply comoves with the Hubble flow, consistent with Eq. (5.5). The correction term encodes the growing perturbation. The mean density within the shell, $\rho = 3M/(4\pi r^3)$, with $M = \pi^2 r_{\max}^3/(8Gt_{\max}^2)$ from Eq. (5.3), is

$$\rho(t) = \frac{1}{6\pi G t^2} \left[1 + \frac{3}{20} \left(\frac{6\pi t}{t_{\max}}\right)^{2/3} + \dots\right], \quad (5.10)$$

whose leading term is exactly the EdS background density. The mean overdensity within the shell therefore grows at early times as

$$\delta(t) = \frac{3}{20} \left(\frac{6\pi t}{t_{\max}}\right)^{2/3} = \frac{3}{5} \delta_i \left(\frac{t}{t_i}\right)^{2/3} \propto a, \quad (5.11)$$

where the second equality uses Eq. (5.7). We recover the linear growing mode $D_+ \propto a$ found in Sec. 4.2.

Eq. (5.11) lets us compare the true non-linear evolution with its linear extrapolation. At turnaround, the linear theory predicts $\delta_{\text{lin}}(t_{\max}) = (3/20)(6\pi)^{2/3} \simeq 1.06$, while the true mean overdensity is $1 + \delta = 9\pi^2/16 \simeq 5.6$. At collapse, $t_{\text{col}} = 2t_{\max}$, the true density formally diverges, while the linear theory predicts

$$\delta_c \equiv \delta_{\text{lin}}(t_{\text{col}}) = \frac{3}{20} (12\pi)^{2/3} \simeq 1.686. \quad (5.12)$$

The number $\delta_c \simeq 1.686$ is the workhorse of halo statistics: a region collapses into a halo once its *linearly extrapolated* overdensity exceeds δ_c . We will use it in the Press-Schechter formalism of Sec. 5.2.

Exercise 5.1 (turnaround relations for a perturbed Hubble flow). Linear perturbation theory tells us that a pure growing mode comes with a slightly perturbed initial velocity,

$$\left. \frac{dr}{dt} \right|_{t_i} = \frac{2r_i}{3t_i} \left(1 - \frac{\delta_i}{3} \right). \quad (5.13)$$

Repeat the derivation of Eq. (5.7) with this initial condition and show that

$$r_{\max} = \frac{3}{5} \frac{r_i}{\delta_i}, \quad t_{\max} = \frac{3\pi}{4} t_i \left(\frac{5\delta_i}{3} \right)^{-3/2}, \quad (5.14)$$

i.e., the results of Eq. (5.7) with the replacement $\delta_i \rightarrow 5\delta_i/3$. Verify that Eq. (5.11) now reads $\delta(t) = \delta_i (t/t_i)^{2/3}$: the entire initial perturbation resides in the growing mode.

Of course, the shell does not really collapse to a point: small deviations from spherical symmetry convert the radial infall into random motions, and the system settles into virial equilibrium. This process is called virialization.

Question 5.1. Why is the halo overdensity $\Delta \approx 200$ special?

Answer. Model the collapsing region as a uniform sphere of mass M , whose gravitational potential energy at radius r is $U(r) = -\frac{3}{5}GM^2/r$. At turnaround, the sphere is momentarily at rest, so its total energy is purely potential,

$$E = U(r_{\max}) = -\frac{3}{5} \frac{GM^2}{r_{\max}}. \quad (5.15)$$

After collapse, the sphere settles into virial equilibrium, where the virial theorem imposes $U = -2K$ and hence $E = U + K = U/2$. Energy conservation between the two moments fixes the virial radius:

$$U(r_{\max}) = \frac{1}{2}U(r_{\text{vir}}) \quad \Rightarrow \quad r_{\text{vir}} = \frac{r_{\max}}{2}. \quad (5.16)$$

The standard bookkeeping takes virialization to complete at the collapse time, $\theta = 2\pi$, i.e., $t_{\text{col}} = 2t_{\max}$. It remains to compute the two densities. Solving the last relation of Eq. (5.3) for the enclosed mass gives $M = \pi^2 r_{\max}^3 / (8Gt_{\max}^2)$, so the mean density of the virialized halo is

$$\rho_{\text{halo}} = \frac{M}{\frac{4\pi}{3}r_{\text{vir}}^3} = \frac{8M}{\frac{4\pi}{3}r_{\max}^3} = \frac{3\pi}{4Gt_{\max}^2}, \quad (5.17)$$

while the EdS background density at the collapse time is

$$\rho_b(t_{\text{col}}) = \frac{1}{6\pi G (2t_{\max})^2} = \frac{1}{24\pi G t_{\max}^2}. \quad (5.18)$$

Taking the ratio, all dimensionful quantities drop out:

$$\Delta_{\text{vir}} = \frac{\rho_{\text{halo}}}{\rho_b(t_{\text{col}})} = 18\pi^2 \simeq 178. \quad (5.19)$$

An alternative bookkeeping, which defines the virial mass as the mass within the half-mass radius of the relaxed (King-like) profile, gives ≈ 200 . Both estimates motivate the customary threshold $\Delta = 200$ used in $M_{200\text{m}}$ and $R_{200\text{m}}$, where “m” refers to the mean *matter* density.

5.1.2 Secondary infall and self-similar solutions*

A real halo does not form in a single collapse; it keeps accreting the surrounding mass shells. The classic framework for this *secondary infall* is the self-similar solution of Fillmore and Goldreich [101] and Bertschinger [102]. Consider a scale-free initial perturbation in an EdS universe,

$$\delta_i \equiv \frac{\delta M_i}{M_i} = \left(\frac{M_i}{M_0} \right)^{-1/s}, \quad s > 0, \quad (5.20)$$

where M_i is the initial mass enclosed by a shell and M_0 is a reference mass. A larger s corresponds to a shallower initial perturbation. Since the perturbation decreases outward, inner shells turn around earlier; each shell obeys the single-shell relations Eq. (5.7) with its own $\delta_i(M_i)$. Asking, at cosmic time t , which shell is currently turning around, Eq. (5.7) yields the current turnaround radius and mass,

$$r_{\text{ta}}(t) \propto t^{\frac{2s+6}{9}}, \quad M_{\text{ta}}(t) \propto t^{\frac{2s}{3}}, \quad (5.21)$$

and, before any shell crossing, the enclosed mass profile

$$M(r, t) \propto r^{\frac{3s}{3+s}} \quad \text{for } r < r_{\text{ta}}(t). \quad (5.22)$$

After turnaround, shells collapse, oscillate through the center, and cross one another, so the mass enclosed by a given shell is no longer constant. The problem appears circular: the mass profile determines the trajectories, while the trajectories build up the mass profile. The self-similar ansatz cuts the knot. With the initial condition Eq. (5.20) in an EdS universe, t is the only independent scale, so the profiles must take the form

$$M(r, t) = M_{\text{ta}}(t) \mathcal{M} \left(\frac{r}{r_{\text{ta}}(t)} \right), \quad \rho(r, t) = \rho_b(t) \mathcal{D} \left(\frac{r}{r_{\text{ta}}(t)} \right), \quad (5.23)$$

where $\mathcal{M}$ and $\mathcal{D} = \frac{1}{3}(r/r_{\text{ta}})^{-2} d\mathcal{M}/d(r/r_{\text{ta}})$ are dimensionless functions, and all shells follow a single dimensionless trajectory $\lambda(\tau)$, with $\lambda \equiv r/r_{\text{max}}$ and $\tau \equiv t/t_{\text{max}}$ referring to each shell’s own turnaround scales. The equation of motion Eq. (5.1) becomes

$$\frac{d^2\lambda}{d\tau^2} = -\frac{\pi^2}{8} \frac{\tau^{2s/3}}{\lambda^2} \mathcal{M} \left(\frac{\lambda}{\tau^{\frac{2}{3} + \frac{2s}{9}}} \right), \quad \lambda(1) = 1, \quad \left. \frac{d\lambda}{d\tau} \right|_{\tau=1} = 0, \quad (5.24)$$

where $\mathcal{M}$ is reconstructed from $\lambda(\tau)$ itself by counting the shells currently inside radius r . Solving the coupled system iteratively yields the trajectories and density profiles shown

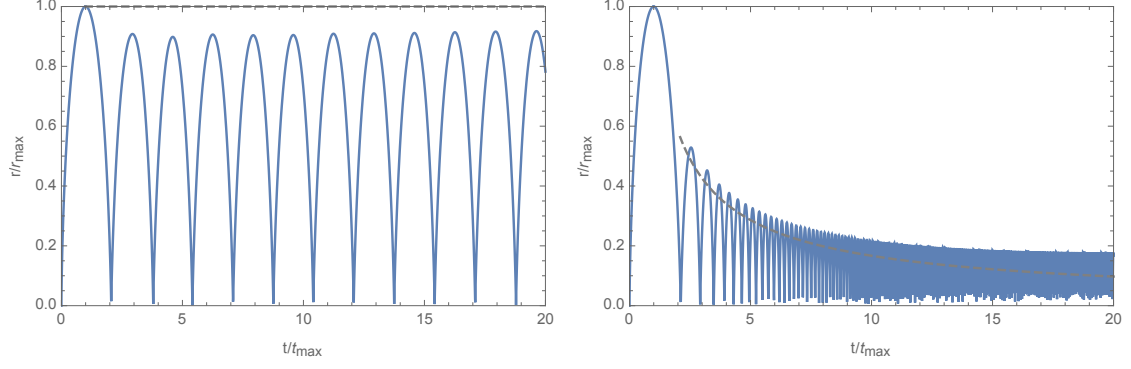


Figure 18. Self-similar shell trajectories $r/r_{\max}$ as functions of $t/t_{\max}$ for $s = 1$ (left) and $s = 5$ (right). For $s = 1 < 3/2$, the apocenter remains an approximately constant fraction of $r_{\max}$; the dashed line marks $r = r_{\max}$. For $s = 5 > 3/2$, the apocenter contracts as the halo keeps growing; the dashed curve shows the envelope $\propto t^q$ with $q = (3 - 2s)/9$.

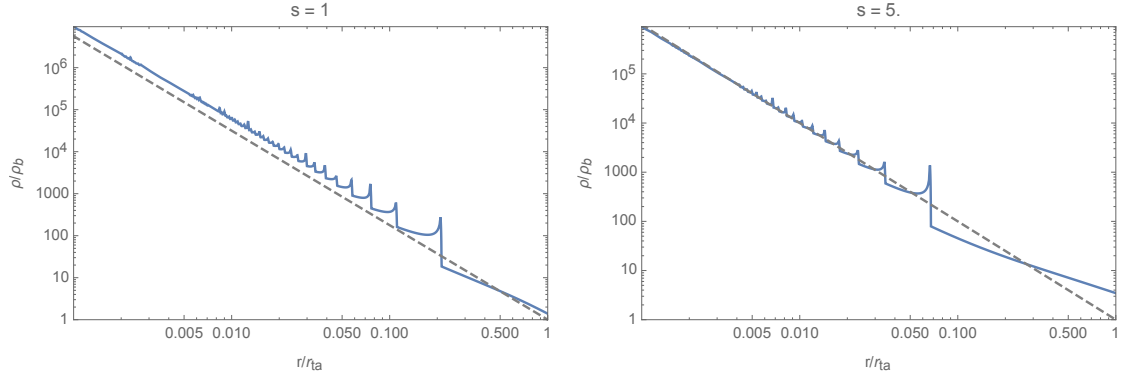


Figure 19. Self-similar density profiles, in units of the background density, for $s = 1$ (left) and $s = 5$ (right). The dashed lines show the asymptotic slopes in Eq. (5.25). The spikes are caustics; the outermost caustic marks the splashback radius.

in Figs. 18 and 19. The density profile exhibits a series of spikes: these are *caustics*, where shells momentarily pile up at their apocenters.

The asymptotic inner profiles can be obtained analytically. Parameterize the apocenter radius of a shell as $r_a/r_{\max} = (t/t_{\max})^q$ and the inner mass profile as $M(r, t) \propto r^\Upsilon$, so that $\rho \propto r^{\Upsilon-3}$. Using the adiabatic invariance of the radial action for orbits much shorter than the halo growth time, Fillmore and Goldreich found [101]

$$\begin{cases} \Upsilon = \frac{3s}{3+s}, & q = 0, & \rho \propto r^{-\frac{9}{3+s}} & \text{for } s \leq \frac{3}{2}, \\ \Upsilon = 1, & q = \frac{3-2s}{9} \leq 0, & \rho \propto r^{-2} & \text{for } s \geq \frac{3}{2}. \end{cases} \quad (5.25)$$

The two regimes have an intuitive explanation in terms of the specific binding energy of a shell at its turnaround,

$$E(M_i) \propto -\frac{GM_i}{r_{\max}} \propto -\frac{M_i \delta_i}{r_i} \propto -M_i^{\frac{2}{3}-\frac{1}{s}}. \quad (5.26)$$

For $s < 3/2$, smaller shells are more tightly bound: the inner potential is dominated by the shells that collapsed long ago and is effectively static (“frozen”), so apocenters do not evolve ($q = 0$), and the inner mass profile simply records the turnaround mass, $M(r) \propto r^{3s/(3+s)}$, as in Eq. (5.22). For $s > 3/2$, smaller shells are more loosely bound: the potential keeps deepening as new mass piles up, previously accreted shells lose energy, their apocenters contract ($q < 0$), and the profile self-adjusts to the isothermal form $\rho \propto r^{-2}$.

5.1.3 The splashback radius*

Where does a halo end? The density profiles in Fig. 19 suggest a natural answer. The *splashback radius*, R_{sp} , is the radius at which the most recently accreted material reaches its first apocenter after turnaround. It coincides with the outermost density caustic, separates the infalling (single-stream) region from the orbiting (multi-stream) region, and manifests itself as a sharp steepening of the logarithmic density slope, which is how it is located in simulations and observations [103–105]. Unlike density-threshold definitions such as $R_{200\text{m}}$, it tracks the actual orbits of the accreted matter: it is free of *pseudo-evolution* (the growth of $M_{200\text{m}}$ caused merely by the dilution of the reference density $200\bar{\rho}_{\text{m}}$) and includes the mass being accreted at the halo boundary. In the frozen regime ($s \leq 3/2$), a shell turns around at t_{max} , crosses the center at $2t_{\text{max}}$, and reaches its first apocenter near r_{max} at $t_{\text{sp}} \approx 3t_{\text{max}}$, so

$$\frac{R_{\text{sp}}}{r_{\text{ta}}(t_{\text{sp}})} \approx \left(\frac{t_{\text{sp}}}{t_{\text{max}}} \right)^{-\frac{2}{3} - \frac{2s}{9}} \approx 3^{-\frac{2}{3} - \frac{2s}{9}}, \quad (5.27)$$

about 0.4 for $s = 1$, in reasonable agreement with the outermost caustic in Fig. 19.

5.1.4 The NFW profile and the Eddington inversion

The self-similar solutions predict pure power-law halos. Cosmological N-body simulations, which follow hierarchical growth with realistic initial conditions, mergers, and non-radial motions, instead find that CDM halos of all masses relax to a universal, gently curving profile, the Navarro-Frenk-White (NFW) profile [106, 107],

$$\rho(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}, \quad (5.28)$$

where ρ_s and r_s are the scale density and the scale radius. The logarithmic slope runs from -1 inside r_s , a cusp milder than the self-similar slopes of Eq. (5.25), to -3 outside. The enclosed mass and the gravitational potential are

$$M(r) = 4\pi\rho_s r_s^3 \left[\ln \left(1 + \frac{r}{r_s} \right) - \frac{r/r_s}{1 + r/r_s} \right], \quad \Phi(r) = -4\pi G \rho_s r_s^3 \frac{\ln(1 + r/r_s)}{r}. \quad (5.29)$$

Since $M(r)$ grows logarithmically at large radii, an NFW halo must be truncated at a boundary, e.g., $R_{200\text{m}}$ or the splashback radius. The two NFW parameters are often traded for a halo mass, e.g., $M_{200\text{m}}$, and the concentration, $c \equiv R_{200\text{m}}/r_s$. Smaller halos collapse earlier, when the universe was denser, and end up more concentrated; the resulting concentration-mass-redshift relation encodes the assembly history of halos.

For many applications, the density profile alone is not enough. The event rate in a direct detection experiment depends on the local *velocity* distribution of dark matter; the scattering rate in an SIDM halo depends on the distribution of relative velocities. What we need is the phase-space distribution function (DF), $f(\mathbf{r}, \mathbf{v})$. For a spherical, isotropic, self-gravitating system in equilibrium, the Jeans theorem restricts the DF to depend on the phase-space coordinates only through the relative energy per unit mass,

$$f(\mathbf{r}, \mathbf{v}) = f(\mathcal{E}), \quad \mathcal{E} = \Psi(r) - \frac{v^2}{2}, \quad (5.30)$$

where $\Psi \equiv -\Phi + \Phi_0$ is the relative potential, with the constant Φ_0 chosen such that Ψ vanishes at the halo boundary. Bound particles have $0 < \mathcal{E} \leq \Psi(r)$, i.e., $v < v_{\text{esc}}(r) = \sqrt{2\Psi(r)}$. The density is recovered by integrating over velocities,

$$\rho(r) = \int d^3v f(\mathcal{E}) = 4\pi \int_0^{\sqrt{2\Psi}} dv v^2 f\left(\Psi - \frac{v^2}{2}\right) = 4\pi\sqrt{2} \int_0^\Psi d\mathcal{E} f(\mathcal{E}) \sqrt{\Psi - \mathcal{E}}. \quad (5.31)$$

In practice, we face the inverse problem: given a density profile, inferred from observations or simulations, what DF generates it? Since $\Psi(r)$ decreases monotonically outward, we can regard ρ as a function of Ψ . Eq. (5.31) then becomes an Abel integral equation for f , whose solution (Exercise 5.2) is *Eddington's formula* [108],

$$f(\mathcal{E}) = \frac{1}{2\sqrt{2}\pi^2} \frac{d}{d\mathcal{E}} \int_0^\mathcal{E} \frac{d\Psi}{\sqrt{\mathcal{E} - \Psi}} \frac{d\rho}{d\Psi}. \quad (5.32)$$

Fed with any spherical density profile, e.g., the NFW profile with the potential in Eq. (5.29), the Eddington inversion delivers the full isotropic phase-space structure of the halo. It is widely used to set up equilibrium halos as initial conditions for N-body experiments and to model the dark matter velocity distribution of the Milky Way, providing a self-consistent alternative to the Maxwellian ansatz adopted in standard direct detection analyses. Two caveats are worth remembering: Eq. (5.32) does not guarantee $f \geq 0$, so a density-potential pair that yields a negative f simply cannot be supported by an isotropic DF; and real halos are neither perfectly spherical nor isotropic, in which case anisotropic generalizations exist (see [109]).

Exercise 5.2 (Eddington inversion). Solve the Abel integral equation

$$h(x) = \int_0^x \frac{\phi(s)}{\sqrt{x-s}} ds \quad (5.33)$$

for $\phi(s)$, showing that

$$\phi(s) = \frac{1}{\pi} \frac{d}{ds} \int_0^s \frac{h(x)}{\sqrt{s-x}} dx. \quad (5.34)$$

Hint: Insert h into the claimed solution, exchange the order of the two integrations, and use $\int_{s'}^s dx [(x-s')(s-x)]^{-1/2} = \pi$. Then derive Eq. (5.32): differentiate Eq. (5.31)

once with respect to Ψ to bring it to the Abel form, and apply the inversion with $x \rightarrow \Psi$ and $s \rightarrow \mathcal{E}$.

5.2 Hierarchical structure formation

The spherical collapse model tells us *when* a region collapses: once its linearly extrapolated overdensity exceeds $\delta_c \simeq 1.686$. To turn this into a census of halos, we need the *halo mass function*, $n(M, t)$, the comoving number density of halos per unit mass. The Press-Schechter (PS) formalism [110] obtains it by combining the collapse criterion of Sec. 5.1 with the Gaussian statistics of the initial density field.⁶

5.2.1 The smoothed density field

In linear theory, the density field factorizes as $\delta_m(\mathbf{x}, t) = D_+(t) \delta_0(\mathbf{x})$, where δ_0 is the field linearly extrapolated to today and the growth factor is normalized to $D_+(t_0) = 1$ (in an EdS universe, $D_+ \propto a$, see Sec. 4.2). The collapse criterion then admits two equivalent readings: a region at $\mathbf{x}$ has collapsed by time t if $\delta_m(\mathbf{x}, t) > \delta_c$, or, keeping the field static and letting the threshold evolve, if

$$\delta_0(\mathbf{x}) > \delta_c(t) \equiv \frac{1.686}{D_+(t)}. \quad (5.35)$$

The barrier $\delta_c(t)$ decreases with time: an ever larger fraction of space collapses, and larger and larger halos form, in line with hierarchical structure formation.

Eq. (5.35) contains no notion of the halo *mass* yet. To introduce one, we smooth δ_0 over a comoving radius R with a (top-hat) window function W ,

$$\delta_M(\mathbf{x}) \equiv \int d^3x' \delta_0(\mathbf{x}') W(|\mathbf{x} - \mathbf{x}'|; R), \quad M = \frac{4\pi}{3} \bar{\rho}_m R^3, \quad (5.36)$$

where $\bar{\rho}_m$ is the comoving mean matter density. A region of δ_M crossing the barrier corresponds to the collapse of a mass M . Being a linear functional of a Gaussian field, δ_M is itself Gaussian, with zero mean and variance

$$\sigma^2(M) = \langle \delta_M^2 \rangle = \int d \ln k \Delta^2(k) \widetilde{W}^2(kR), \quad \widetilde{W}(kR) = \frac{3 [\sin(kR) - kR \cos(kR)]}{(kR)^3}, \quad (5.37)$$

where $\Delta^2(k)$ is the dimensionless linear matter power spectrum of Sec. 4.2, extrapolated to today, and $\widetilde{W}$ is the Fourier transform of the top-hat window. The quantity $\sigma(M)$ is called the mass variance.⁷ For a CDM-like spectrum, $\sigma(M)$ decreases monotonically with M : small scales are the most overdense and collapse first.

⁶This subsection is based on Frank van den Bosch's ASTR 610 lecture notes (Lecture 9: Press-Schechter Theory) and Wayne Hu's Astro 321 lecture notes (Set 7).

⁷The measured normalization of the power spectrum is conventionally quoted as $\sigma_8 \equiv \sigma(R = 8 h^{-1} \text{ Mpc}) \simeq 0.8$.

5.2.2 The Press-Schechter mass function

Press and Schechter postulated that *the fraction of mass residing in halos with masses greater than M at time t equals twice the probability that δ_M exceeds the barrier*:

$$F(> M, t) = 2 \mathcal{P}[\delta_M > \delta_c(t)] = \frac{2}{\sqrt{2\pi} \sigma(M)} \int_{\delta_c(t)}^{\infty} d\delta e^{-\frac{\delta^2}{2\sigma^2(M)}} = \operatorname{erfc}\left(\frac{\nu}{\sqrt{2}}\right), \quad \nu \equiv \frac{\delta_c(t)}{\sigma(M)}. \quad (5.38)$$

The factor of 2 is the notorious PS “fudge factor.” Without it, at most half of the mass could ever end up in halos ($\mathcal{P} \rightarrow 1/2$ as $\sigma \rightarrow \infty$), because only initially *overdense* points are counted. However, an underdense point embedded in a larger-scale overdense region does end up inside a halo. This is the “cloud-in-cloud” problem, which also plagues the alternative approach of counting the peaks of the smoothed field [111]. Press and Schechter simply multiplied by two by hand; the excursion set derivation shows that this is exactly right.

The mass function follows by differentiation: $(\partial F / \partial M) dM$ is the mass fraction in halos with masses in $[M, M + dM]$, and multiplying by $\bar{\rho}_m / M$ converts it into a number density,

$$n(M, t) dM = \frac{\bar{\rho}_m}{M} \left| \frac{\partial F}{\partial M} \right| dM = \sqrt{\frac{2}{\pi}} \frac{\bar{\rho}_m}{M^2} \nu e^{-\nu^2/2} \left| \frac{d \ln \sigma}{d \ln M} \right| dM. \quad (5.39)$$

The shape of Eq. (5.39) is controlled by the characteristic mass $M^*(t)$, defined by $\sigma(M^*) = \delta_c(t)$, i.e., $\nu(M^*) = 1$; today, $M^* \sim 10^{13} h^{-1} M_\odot$. Above M^* , halos are exponentially rare; below M^* , the mass function rises as a power law. Since $\delta_c(t)$ decreases with time, $M^*(t)$ grows: increasingly massive halos form as time goes on, the statistical counterpart of the merger tree in Fig. 16.

Exercise 5.3 (PS mass function). Derive Eq. (5.39) from Eq. (5.38). For a power-law spectrum, $P_m(k) \propto k^n$ with $n > -3$, show that $\sigma(M) \propto M^{-(n+3)/6}$ and hence $n(M) \propto M^{(n-9)/6}$ for $M \ll M^*$. For CDM, $n \rightarrow -3$ on small scales, so the mass function diverges toward low masses as $n(M) \propto M^{-2}$.

Despite its crude assumptions, the PS mass function agrees with N-body simulations to within a factor of a few; the residuals (too few massive halos, too many small ones) are largely cured by replacing the constant barrier with the mass-dependent barrier of ellipsoidal collapse [112]. For dark matter physics, the mass function is the bridge between the linear power spectrum and halo counts: a beyond-CDM suppression of $\Delta^2(k)$ propagates through $\sigma(M)$ into a deficit of small halos, which underlies the satellite-count limits in Tab. 3.

5.3 Post formation evolution of dark matter halo: self-interacting dark matter case

CDM has encountered several challenges at small scales, including the core vs. cusp problem, the diversity problem, and the too-big-to-fail problem (see review by [113]). SIDM offers a common solution by providing an intrinsic mechanism to alter the dark matter distribution within halos. The core formed during the early gravothermal evolution in SIDM

can address both the core vs. cusp and too-big-to-fail problems. Additionally, the super-cuspy profile that develops in the late core-collapse stage, along with the early-stage core, may resolve the full scope of the diversity problem.

5.3.1 A qualitative understanding

The behavior of self-interacting dark matter (SIDM) at different stages can be understood, at the zeroth order, using the hydrostatic equation:

$$\frac{\nabla P}{\rho} = -\nabla\Phi, \quad (5.40)$$

where $P \equiv \rho\nu^2$ is the SIDM pressure, ν is the one-dimensional velocity dispersion, and Φ is the gravitational potential of the halo. Under spherical symmetry, this equation can be rewritten as:

$$\frac{\partial(\rho\nu^2)}{\partial r} = -\frac{GM(<r)\rho}{r^2}, \quad (5.41)$$

where $M(<r)$ is the enclosed mass. Integrating both left and right hand side by r , we have

$$-\frac{1}{2}\rho\nu^2 = -\frac{1}{2}\int_r^\infty dr' \frac{GM\rho}{r'^2} \Rightarrow K.E. = -\frac{1}{2}P.E., \quad (5.42)$$

which is just the Virial theorem.

Let us consider the density profile at the time of halo formation. Since the dynamical time of a halo is typically much shorter than the SIDM relaxation time, the density profile at halo formation will be similar to that of a CDM halo, i.e., the NFW profile of Eq. (5.28). For the inner halo region, the density scales as $\rho \propto r^{-1}$. Therefore, $M \propto r^2$. Assuming $\nu \propto r^s$, the left-hand side (LHS) and right-hand side (RHS) of Eq. (5.41) respectively scale as:

$$\text{LHS} \propto \frac{r^{-1}r^{2s}}{r} = r^{2s-2}, \quad \text{RHS} \propto \frac{r^2r^{-1}}{r^2} = r^{-1}. \quad (5.43)$$

Requiring the LHS and RHS to have the same scaling behavior gives:

$$2s - 2 = -1 \quad \Rightarrow \quad s = \frac{1}{2}. \quad (5.44)$$

$\nu \propto r^{1/2}$ means the halo will be cold at the center and hot at the scale radius. Such an initial temperature imbalance creates an opportunity for SIDM to manifest its effect: thermalizing regions with different temperatures.

Next, let us consider the end result of the self-interactions, the isothermal limit, where ν is independent of r . In this case, we can move ν^2 out of the partial derivative on the LHS of Eq. (5.41) and rewrite the equation as:

$$\frac{\partial \ln \rho}{\partial \ln r} = -\frac{GM(<r)}{r\nu^2}. \quad (5.45)$$

There are two scenarios to consider:

1. **Finite central density:** $\lim_{r \rightarrow 0} \rho = \rho_0$. In this case, $\lim_{r \rightarrow 0} M = \frac{4\pi}{3}\rho_0 r^3$ and ν is independent of r , therefore $\lim_{r \rightarrow 0} \text{RHS} = 0$. This leads to $\lim_{r \rightarrow 0} \frac{\partial \ln \rho}{\partial \ln r} = 0$, which means the isothermal profile can have a flat core.

2. **Singular central density:** we assume the density follows the ansatz $\rho = Ar^x$. Substituting it into Eq. (5.45) gives:

$$x = -\frac{4\pi G}{3}Ar^{x+2}\nu^{-2} \Rightarrow x = -2, A = \frac{3\nu^2}{2\pi G} \Rightarrow \rho = \frac{3\nu^2}{2\pi Gr^2}. \quad (5.46)$$

This is the so-called singular isothermal profile. Note that this is one solution but not the only solution of Eq. (5.45).

Therefore, an SIDM halo, which obeys Eq. (5.45), is capable of producing both a cored profile and a singular isothermal profile. More detailed analysis shows that the density profile in the inner part of a core-collapsed halo (outside the secondary core) scales as $r^{-2.2}$. The slight deviation from r^{-2} is significant because the density slope is related to the velocity dispersion slope by

$$\frac{d \ln \nu}{d \ln r} = 1 + \frac{1}{2} \frac{d \ln \rho}{d \ln r}, \quad (5.47)$$

which can again be derived from the hydrostatic equation (see Exercise 5.4). Therefore,

$$\frac{d \ln \rho}{d \ln r} < -2 \Rightarrow \frac{d \ln \nu}{d \ln r} < 0. \quad (5.48)$$

A negative velocity dispersion gradient means heat flows out from the core of the halo to its outskirts. The process is necessary for SIDM to evolve from a cored phase to a core-collapsed phase. Unlike typical thermal systems, an SIDM halo, viewed as a gravothermal system, has a negative heat capacity. This can be seen from the Virial theorem, where

$$2K.E. + P.E. = 0 \Rightarrow \text{Total } E = -K.E. \Rightarrow C = \frac{\text{Total } E}{T} < 0, \quad (5.49)$$

where in the last step we assume $K.E. \propto T$. For a negative heat capacity system, the heat outflow *heats* the core and generates more heat outflow, and the system's maximum entropy is infinite, leading to a core-collapsed halo in a self-similar way. Another classical negative heat capacity system is the black hole. As a black hole radiates more Hawking radiation, its temperature gets increased.

Exercise 5.4 (velocity dispersion slope). Starting from the hydrostatic equation Eq. (5.41), derive the relation between the velocity dispersion slope and the density slope, $\frac{d \ln \nu}{d \ln r} = 1 + \frac{1}{2} \frac{d \ln \rho}{d \ln r}$.

Hint: Assume power-law behaviors $\rho \propto r^a$ and $\nu \propto r^s$ in the inner halo and match the scaling of the two sides of Eq. (5.41).

The evolution time from the initial NFW profile to the cored profile and then to the quasi-isothermal profile depends on the relaxation time of dark matter self-interaction. The relaxation time is typically given by

$$t_{\text{rlx}} = \frac{1}{n\sigma v} = \frac{1}{\rho(\sigma/m)v} = 1.5 \text{ Gyr} \left(\frac{0.4 \text{ GeV cm}^{-3}}{\rho} \right) \left(\frac{1 \text{ cm}^2 \text{ g}^{-1}}{\sigma/m} \right) \left(\frac{300 \text{ km s}^{-1}}{v_{\text{rel}}} \right). \quad (5.50)$$

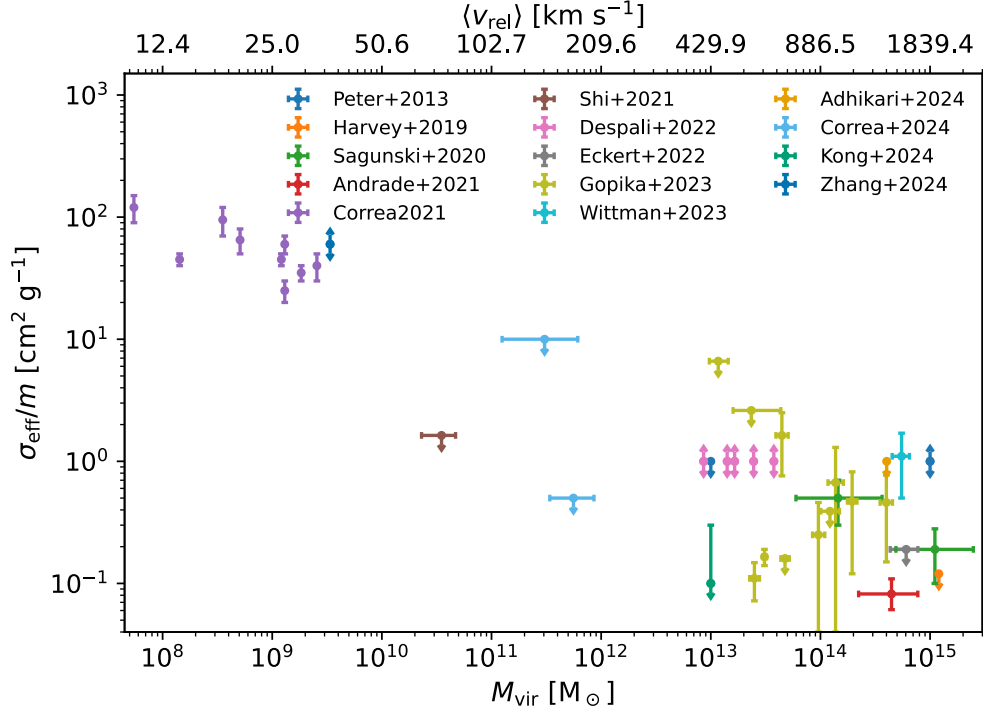


Figure 20. The allowed effective SIDM cross-section strength as a function of relative velocity of incoming dark matter particles or typical halo virial mass. Sources and references: [114]. The probes can be mainly classified into four types: (1) halo crossing [115–122] (2) core shape [123, 124] (3) core density and density slope [125, 126], and (4) subhalo evaporation [127, 128].

Simulations show that it takes $t = \mathcal{O}(1) \times t_{\text{rlx}}$ for the halo to evolve from the initial NFW profile to the cored profile, and $t = \mathcal{O}(100) \times t_{\text{rlx}}$ for the halo to further evolve to the quasi-isothermal profile, which is the moment when a significant fraction of the halo has evaporated. Note that this evolution time is strongly influenced by the halo properties, in particular the halo concentration c_h , defined as the halo radius r_h over the scale radius r_s of the initial halo profile. In summary, SIDM halos can exhibit NFW, cored, or quasi-isothermal profiles depending on their evolution time and halo properties.

5.3.2 Astrophysical probes and the velocity-dependent SIDM*

There are many astrophysical probes of SIDM for different halo systems. We summarize the *allowed* values from these probes and the sweet spots for solving small-scale challenges in Fig. 20 as colored bars. The probes fall mostly into four classes:

1. Halo crossing: SIDM leads to smaller separations between the crossed clusters.
2. Core shape: SIDM leads to a more spherical halo core.
3. Core density and density slope: SIDM halos can have a lower density with a flat density slope (core expansion phase) or a higher density with a cuspy density slope

(core collapse phase). Such information can be inferred from gravitational rotation curves, strong lensing, weak lensing, or gravitational wave lensing.

4. Subhalo evaporation: SIDM subhalos are more likely to be heated by the host SIDM halo (another type of “core expansion”) and thus evaporate.

See the references in the caption. Ideally, if we can observe the SIDM signatures across all types of probes, this will make a strong case for SIDM. It is also clear that a velocity-dependent cross-section strength is necessary for SIDM to evade observational constraints and explain the small-scale challenges.

The velocity dependence of SIDM can be effectively realized in the so-called Yukawa SIDM model. This model considers a light scalar with mass m_ϕ mediating the long-range force between dark matter particles with mass m_χ . The self-interaction is described by a Yukawa potential:

$$V(r) = \pm \alpha_\chi \frac{e^{-m_\phi r}}{r}, \quad (5.51)$$

where α_χ is the strength of the potential, m_ϕ is the mediator mass, and $+$ or $-$ represents a repulsive or attractive interaction, respectively. Given the differential scattering cross section $d\sigma/d\Omega$ corresponding to $V(r)$, we compute the viscosity transfer cross section as:

$$\sigma_V = \int \frac{d\sigma}{d\Omega} \sin^2 \theta d\Omega, \quad (5.52)$$

where θ is the scattering angle. The $\sin^2 \theta$ kernel appears from deriving the thermal conductivity [129], which is the key quantity driving the gravothermal evolution of a SIDM halo. The parameter space of the model can be divided into three regimes: the perturbative Born regime ($\alpha_\chi m_\chi/m_\phi \ll 1$), the non-perturbative classical regime ($\alpha_\chi m_\chi/m_\phi \gg 1$, $m_\chi v/m_\phi \gg 1$), and the non-perturbative resonant regime ($\alpha_\chi m_\chi/m_\phi \gg 1$, $m_\chi v/m_\phi \ll 1$). This model exhibits a variety of velocity dependencies, as shown in Fig. 21, which is based on the parameters listed in Tab. 4. The computational details can be found in [130]. Basically, one can apply the partial-wave analysis to solve the wave function of dark matter for the scattering process. The viscosity transfer cross section is given by

$$\sigma_V = \frac{4\pi}{k^2} \sum_{\ell=0}^{\ell_{\max}} \frac{(\ell+1)(\ell+2)}{2\ell+3} \sin^2(\delta_{\ell+2} - \delta_\ell), \quad (5.53)$$

where $k = m_\chi v/2$ is the momentum of the dark matter in the center of mass frame. $\ell_{\max}$ is the maximum partial wave number. δ_ℓ is the phase shift. It can be viewed as the large radius limit of an auxiliary function $\delta_\ell(r)$, $\delta_\ell = \lim_{r \rightarrow \infty} \delta_\ell(r)$. $\delta_\ell(r)$ can be obtained by solving

$$\frac{\partial \delta_\ell(r)}{\partial r} = -km_\chi r^2 V(r) (j_\ell(kr) \sin(\delta_\ell(r)) + n_\ell(kr) \cos(\delta_\ell(r)))^2, \quad (5.54)$$

$$\delta_\ell(0) = 0, \quad (5.55)$$

	m_χ [GeV]	m_ϕ [MeV]	α_χ
Model 1	119	0.4	3.0×10^{-3}
Model 2	20	3.5	1.6×10^{-3}
Model 3	40	1.1	1.5×10^{-3}
Model 4	18	1.7	1.2×10^{-3}
Model 5	27	1.1	9.0×10^{-4}

Table 4. Benchmarks for the Yukawa SIDM model shown in Fig. 21.

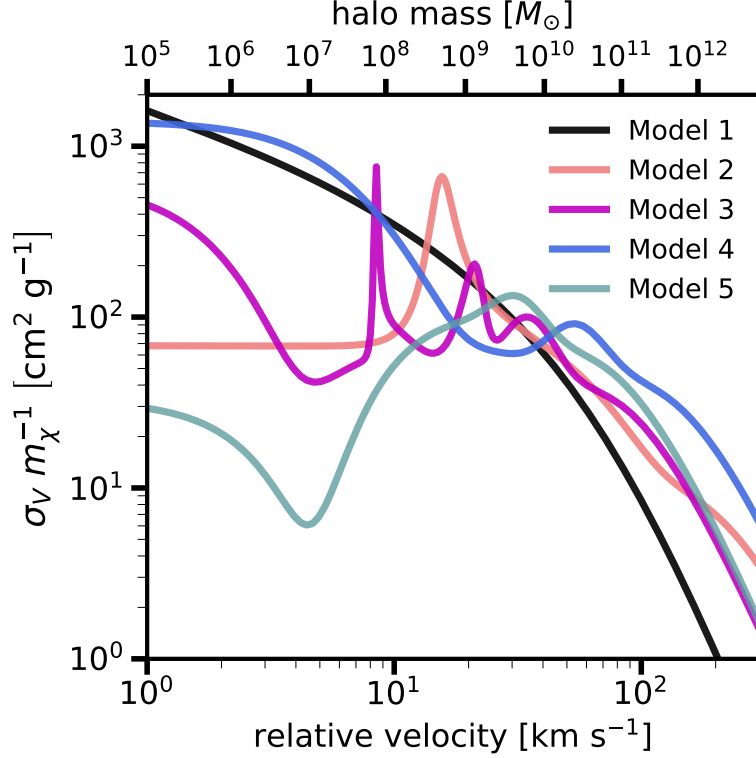


Figure 21. The viscosity transfer cross-section strength σ_V/m_χ for the five benchmark values of a Yukawa SIDM model, plotted as a function of the relative velocity v (lower x -axis). The upper x -axis shows the halo mass with maximum circular velocity $v_{\text{max}} = v$. The plot is taken from [130].

where j_ℓ and n_ℓ are spherical Bessel functions. Note that Yukawa SIDM in small halos generally shows a large cross section strength, which could push dwarf halos toward a core-collapsed phase.

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